

Markov Chains

The main object of study in this chapter is (temporally homogeneous) Markov chains. These processes are important because the assumptions are satisfied in many examples and lead to a rich and detailed theory.

5.1 Examples

We begin with the case of countable state space. In this situation the Markov property is that for any states $i_0, \dots, i_{n-1}, i$, and j

$$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j | X_n = i)$$

In words, given the present state the rest of the past is irrelevant for predicting the future. In this section we will introduce a number of examples. In each case, we will not check the Markov property, but simply give the transition probability

$$p(i, j) = P(X_{n+1} = j | X_n = i)$$

and leave the rest to the reader. In the next section we will be more formal.

Example 5.1.1 (Random walk) Let $\xi_1, \xi_2, \dots \in \mathbf{Z}^d$ be independent with distribution μ . Let $X_n = X_0 + \xi_1 + \dots + \xi_n$, where X_0 is constant. Then X_n is a Markov chain with transition probability.

$$p(i, j) = \mu(\{j - i\})$$

Example 5.1.2 (Branching processes) $S = \{0, 1, 2, \dots\}$

$$p(i, j) = P\left(\sum_{m=1}^i \xi_m = j\right)$$

where $\xi_1, \xi_2, \dots$ are i.i.d. nonnegative integer-valued random variables. In words, each of the i individuals at time n (or in generation n) gives birth to an independent and identically distributed number of offspring.

The first two chains are not good examples because, as we will see, they do not converge to equilibrium.

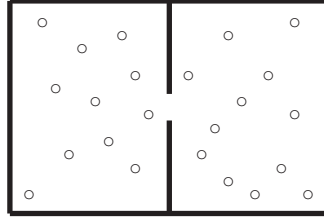


Figure 5.1 Physical motivation for the Ehrenfest chain.

Example 5.1.3 (Ehrenfest chain) $S = \{0, 1, \dots, r\}$

$$\begin{aligned} p(k, k+1) &= (r-k)/r \\ p(k, k-1) &= k/r \\ p(i, j) &= 0 \quad \text{otherwise} \end{aligned}$$

In words, there is a total of r balls in two urns; k in the first and $r-k$ in the second. We pick one of the r balls at random and move it to the other urn. Ehrenfest used this to model the division of air molecules between two chambers (of equal size and shape) that are connected by a small hole.

Example 5.1.4 (Wright–Fisher model) Thinking of a population of $N/2$ diploid individuals who have two copies of each of their chromosomes, or of N haploid individuals who have one copy, we consider a fixed population of N genes that can be one of two types: A or a . In the simplest version of this model the population at time $n+1$ is obtained by drawing with replacement from the population at time n . In this case, if we let X_n be the number of A alleles at time n , then X_n is a Markov chain with transition probability

$$p(i, j) = \binom{N}{j} \left(\frac{i}{N}\right)^j \left(1 - \frac{i}{N}\right)^{N-j}$$

since the right-hand side is the binomial distribution for N independent trials with success probability i/N .

In this model the states $x = 0$ and N that correspond to fixation of the population in the all a or all A states are **absorbing states**, that is, $p(x, x) = 1$. As we will see, this chain will eventually end up in state 0 or state N . To make this simple model more interesting, we introduce mutations. That is, an A that is drawn ends up being an a in the next generation with probability u , while an a that is drawn ends up being an A in the next generation with probability v . In this case, the probability an A is produced by a given draw is

$$\rho_i = \frac{i}{N}(1-u) + \frac{N-i}{N}v$$

but the transition probability still has the binomial form

$$p(i, j) = \binom{N}{j} (\rho_i)^j (1 - \rho_i)^{N-j}$$

If u and v are both positive, then 0 and N are no longer absorbing states, so the system can converge to an equilibrium distribution as time $t \rightarrow \infty$?

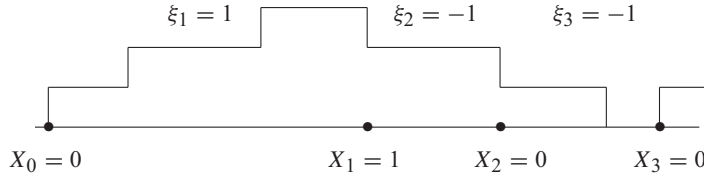


Figure 5.2 Realization of the M/G/1 queue. Black dots indicate the times at which the customers enter service.

Example 5.1.5 (M/G/1 queue) In this model, customers arrive according to a Poisson process with rate λ . (M is for Markov and refers to the fact that in a Poisson process the number of arrivals in disjoint time intervals is independent.) Each customer requires an independent amount of service with distribution F . (G is for general service distribution. 1 indicates that there is one server.) Let X_n be the number of customers waiting in the queue at the time the n th customer enters service. To be precise, when $X_0 = x$, the chain starts with x people waiting in line and customer 0 just beginning her service. To understand the definition, the picture in Figure 5.2 is useful:

To define our Markov chain X_n , let

$$a_k = \int_0^\infty e^{-\lambda t} \frac{(\lambda t)^k}{k!} dF(t)$$

be the probability that k customers arrive during a service time. Let $\xi_1, \xi_2, \dots$ be i.i.d. with $P(\xi_i = k - 1) = a_k$. We think of ξ_i as the net number of customers to arrive during the i th service time, subtracting one for the customer who completed service. We define X_n by

$$X_{n+1} = (X_n + \xi_{n+1})^+ \quad (5.1.1)$$

The positive part only takes effect when $X_n = 0$ and $\xi_{n+1} = -1$ (e.g., $X_2 = 0$, $\xi_3 = -1$ in Figure 5.2) and reflects the fact that when the queue has size 0 and no one arrives during the service time, the next queue size is 0, since we do not start counting until the next customer arrives and then the queue length will be 0.

It is easy to see that the sequence defined in (5.1.1) is a Markov chain with transition probability

$$\begin{aligned} p(0, 0) &= a_0 + a_1 \\ p(j, j - 1 + k) &= a_k \quad \text{if } j \geq 1 \text{ or } k > 1 \end{aligned}$$

The formula for a_k is rather complicated, and its exact form is not important, so we will simplify things by assuming only that $a_k > 0$ for all $k \geq 0$ and $\sum_{k \geq 0} a_k = 1$.

Exercises

- 5.1.1 Let $\xi_1, \xi_2, \dots$ be i.i.d. $\in \{1, 2, \dots, N\}$ and taking each value with probability $1/N$. Show that $X_n = |\{\xi_1, \dots, \xi_n\}|$ is a Markov chain and compute its transition probability.

- 5.1.2 Let $\xi_1, \xi_2, \dots$ be i.i.d. $\in \{-1, 1\}$, taking each value with probability $1/2$. Let $S_0 = 0$, $S_n = \xi_1 + \dots + \xi_n$ and $X_n = \max\{S_m : 0 \leq m \leq n\}$. Show that X_n is not a Markov chain.
- 5.1.3 Let $\xi_0, \xi_1, \dots$ be i.i.d. $\in \{H, T\}$, taking each value with probability $1/2$. Show that $X_n = (\xi_n, \xi_{n+1})$ is a Markov chain and compute its transition probability p . What is p^2 ?
- 5.1.4 **Brother-sister mating.** In this scheme, two animals are mated, and among their direct descendants two individuals of opposite sex are selected at random. These animals are mated and the process continues. Suppose each individual can be one of three genotypes AA, Aa, aa , and suppose that the type of the offspring is determined by selecting a letter from each parent. With these rules, the pair of genotypes in the n th generation is a Markov chain with six states:

$$AA, AA \quad AA, Aa \quad AA, aa \quad Aa, Aa \quad Aa, aa \quad aa, aa$$

Compute its transition probability.

- 5.1.5 **Bernoulli-Laplace model of diffusion.** Suppose two urns, which we will call left and right, have m balls each. b (which we will assume is $\leq m$) balls are black, and $2m - b$ are white. At each time, we pick one ball from each urn and interchange them. Let the state at time n be the number of black balls in the left urn. Compute the transition probability.
- 5.1.6 Let $\theta, U_1, U_2, \dots$ be independent and uniform on $(0, 1)$. Let $X_i = 1$ if $U_i \leq \theta$, $= -1$ if $U_i > \theta$, and let $S_n = X_1 + \dots + X_n$. In words, we first pick θ according to the uniform distribution and then flip a coin with probability θ of heads to generate a random walk. Compute $P(X_{n+1} = 1 | X_1, \dots, X_n)$ and conclude S_n is a temporally inhomogeneous Markov chain. This is due to the fact that “ S_n is a sufficient statistic for estimating θ .”

5.2 Construction, Markov Properties

Let $(S, \mathcal{S})$ be a measurable space. This will be the state space for our Markov chain.

A function $p : S \times \mathcal{S} \rightarrow \mathbf{R}$ is said to be a **transition probability** if:

- (i) For each $x \in S$, $A \rightarrow p(x, A)$ is a probability measure on $(S, \mathcal{S})$.
- (ii) For each $A \in \mathcal{S}$, $x \rightarrow p(x, A)$ is a measurable function.

We say X_n is a Markov chain (w.r.t. $\mathcal{F}_n$) with transition probability p if

$$P(X_{n+1} \in B | \mathcal{F}_n) = p(X_n, B)$$

Given a transition probability p and an **initial distribution** μ on $(S, \mathcal{S})$, we can define a consistent set of finite dimensional distributions by

$$\begin{aligned} P(X_j \in B_j, 0 \leq j \leq n) &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \\ &\quad \cdots \int_{B_n} p(x_{n-1}, dx_n) \end{aligned} \tag{5.2.1}$$

If we suppose that $(S, \mathcal{S})$ is nice, Kolmogorov's extension theorem, Theorem 2.1.21, allows us to construct a probability measure P_μ on **sequence space**

$$(\Omega_\sigma, \mathcal{F}_\infty) = (S^{\{0,1,\dots\}}, \mathcal{S}^{\{0,1,\dots\}})$$

so that the coordinate maps $X_n(\omega) = \omega_n$ have the desired distributions. Note that we have one set of very simple random variables, and a large family of measures, one for each initial condition. Of course, we only need one measure for each state since

$$P_\mu(A) = \int \mu(dx) P_x(A).$$

An advantage of building the chain on this canonical probability space is that we can define the shift operators by

$$\theta_n(\omega_0, \omega_1, \dots) = (\omega_n, \omega_{n+1}, \dots)$$

Our next step is to show

Theorem 5.2.1 X_n is a Markov chain (with respect to $\mathcal{F}_n = \sigma(X_0, X_1, \dots, X_n)$) with transition probability p . That is,

$$P_\mu(X_{n+1} \in B | \mathcal{F}_n) = p(X_n, B)$$

Proof To prove this, we let $A = \{X_0 \in B_0, X_1 \in B_1, \dots, X_n \in B_n\}$, $B_{n+1} = B$, and observe that using the definition of the integral, the definition of A , and the definition of P_μ

$$\begin{aligned} \int_A 1_{(X_{n+1} \in B)} dP_\mu &= P_\mu(A, X_{n+1} \in B) \\ &= P_\mu(X_0 \in B_0, X_1 \in B_1, \dots, X_n \in B_n, X_{n+1} \in B) \\ &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) p(x_n, B_{n+1}) \end{aligned}$$

We would like to assert that the last expression is

$$= \int_A p(X_n, B) dP_\mu$$

To do this, we begin by noting that

$$\int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) 1_C(x_n) = \int_A 1_C(X_n) dP_\mu$$

Linearity implies that for simple functions,

$$\int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) f(x_n) = \int_A f(X_n) dP_\mu$$

and the bounded convergence theorem implies that it is valid for bounded measurable f , e.g., $f(x) = p(x, B_{n+1})$.

The collection of sets for which

$$\int_A 1_{(X_{n+1} \in B)} dP_\mu = \int_A p(X_n, B) dP_\mu$$

holds is a λ -system, and the collection for which it has been proved is a π -system, so it follows from the $\pi - \lambda$ theorem, Theorem 2.1.6, that the equality is true for all $A \in \mathcal{F}_n$. This shows that

$$P(X_{n+1} \in B | \mathcal{F}_n) = p(X_n, B)$$

and proves the desired result. $\square$

Our next small step is to show that if X_n has transition probability p , then for any bounded measurable f

$$E(f(X_{n+1}) | \mathcal{F}_n) = \int p(X_n, dy) f(y) \quad (5.2.2)$$

The desired conclusion is a consequence of the next result, which will save us work in later proofs. Let $\mathcal{H}$ = the collection of bounded functions for which the identity holds.

Theorem 5.2.2 (Monotone class theorem) *Let $\mathcal{A}$ be a π -system that contains Ω and let $\mathcal{H}$ be a collection of real-valued functions that satisfies:*

- (i) *If $A \in \mathcal{A}$, then $1_A \in \mathcal{H}$.*
- (ii) *If $f, g \in \mathcal{H}$, then $f + g$, and $cf \in \mathcal{H}$ for any real number c .*
- (iii) *If $f_n \in \mathcal{H}$ are nonnegative and increase to a bounded function f , then $f \in \mathcal{H}$.*

Then $\mathcal{H}$ contains all bounded functions measurable with respect to $\sigma(\mathcal{A})$.

Proof The assumption $\Omega \in \mathcal{A}$, (ii), and (iii) imply that $\mathcal{G} = \{A : 1_A \in \mathcal{H}\}$ is a λ -system, so by (i) and the $\pi - \lambda$ theorem, Theorem 2.1.6, $\mathcal{G} \supset \sigma(\mathcal{A})$. (ii) implies $\mathcal{H}$ contains all simple functions, and (iii) implies that $\mathcal{H}$ contains all bounded measurable functions. $\square$

To extend (5.2.2), we observe that familiar properties of conditional expectation and (5.2.2) imply

$$\begin{aligned} E\left(\prod_{m=0}^n f_m(X_m)\right) &= E E\left(\prod_{m=0}^n f_m(X_m) \middle| \mathcal{F}_{n-1}\right) \\ &= E\left(\prod_{m=0}^{n-1} f_m(X_m) E(f_n(X_n) | \mathcal{F}_{n-1})\right) \\ &= E\left(\prod_{m=0}^{n-1} f_m(X_m) \int p_{n-1}(X_{n-1}, dy) f_n(y)\right) \end{aligned}$$

The last integral is a bounded measurable function of X_{n-1} , so it follows by induction that if μ is the distribution of X_0 , then

$$\begin{aligned} E\left(\prod_{m=0}^n f_m(X_m)\right) &= \int \mu(dx_0) f_0(x_0) \int p_0(x_0, dx_1) f_1(x_1) \\ &\quad \cdots \int p_{n-1}(x_{n-1}, dx_n) f_n(x_n) \end{aligned} \quad (5.2.3)$$

Our next goal is to prove two extensions of the Markov property in which $\{X_{n+1} \in B\}$ is replaced by a bounded function of the future, $h(X_n, X_{n+1}, \dots)$, and n is replaced by a

stopping time N . These results, especially the second, will be the keys to developing the theory of Markov chains.

Theorem 5.2.3 (The Markov property) *Let $Y : \Omega_o \rightarrow \mathbf{R}$ be bounded and measurable.*

$$E_\mu(Y \circ \theta_m | \mathcal{F}_m) = E_{X_m} Y$$

Remark We denote the function by Y , a letter usually used for random variables, because that's exactly what Y is, a measurable function defined on our probability space Ω_o . Here the subscript μ on the left-hand side indicates that the conditional expectation is taken with respect to P_μ . The right-hand side is the function $\varphi(x) = E_x Y$ evaluated at $x = X_m$.

Proof We begin by proving the result in a special case and then use the $\pi - \lambda$ and monotone class theorems to get the general result. Let $A = \{\omega : \omega_0 \in A_0, \dots, \omega_m \in A_m\}$ and $g_0, \dots, g_n$ be bounded and measurable. Applying (5.2.3) with $f_k = 1_{A_k}$ for $k < m$, $f_m = 1_{A_m} g_0$, and $f_k = g_{k-m}$ for $m < k \leq m+n$ gives

$$\begin{aligned} E_\mu \left(\prod_{k=0}^n g_k(X_{m+k}); A \right) &= \int_{A_0} \mu(dx_0) \int_{A_1} p(x_0, dx_1) \cdots \int_{A_m} p(x_{m-1}, dx_m) \\ &\quad \cdot g_0(x_m) \int p(x_m, dx_{m+1}) g_1(x_{m+1}) \\ &\quad \cdots \int p(x_{m+n-1}, dx_{m+n}) g_n(x_{m+n}) \\ &= E_\mu \left(E_{X_m} \left(\prod_{k=0}^n g_k(X_k) \right); A \right) \end{aligned}$$

The collection of sets for which the last formula holds is a λ -system, and the collection for which it has been proved is a π -system, so using the $\pi - \lambda$ theorem, Theorem 2.1.6, shows that the last identity holds for all $A \in \mathcal{F}_m$.

Fix $A \in \mathcal{F}_m$ and let $\mathcal{H}$ be the collection of bounded measurable Y for which

$$(*) \quad E_\mu(Y \circ \theta_m; A) = E_\mu(E_{X_m} Y; A)$$

The last computation shows that $(*)$ holds when

$$Y(\omega) = \prod_{0 \leq k \leq n} g_k(\omega_k)$$

To finish the proof, we will apply the monotone class theorem, Theorem 5.2.2. Let $\mathcal{A}$ be the collection of sets of the form $\{\omega : \omega_0 \in A_0, \dots, \omega_k \in A_k\}$. $\mathcal{A}$ is a π -system, so taking $g_k = 1_{A_k}$ shows (i) holds. $\mathcal{H}$ clearly has properties (ii) and (iii), so Theorem 5.2.2 implies that $\mathcal{H}$ contains the bounded functions measurable w.r.t $\sigma(\mathcal{A})$, and the proof is complete. $\square$

As corollary of Theorem 5.2.3 we get:

Theorem 5.2.4 (Chapman-Kolmogorov equation)

$$P_x(X_{m+n} = z) = \sum_y P_x(X_m = y) P_y(X_n = z)$$

Intuitively, in order to go from x to z in $m + n$ steps we have to be at some y at time m , and the Markov property implies that for a given y the two parts of the journey are independent.

Proof $P_x(X_{n+m} = z) = E_x(P_x(X_{n+m} = z | \mathcal{F}_m)) = E_x(P_{X_m}(X_n = z))$ by the Markov property, Theorem 5.2.3, since $1_{(X_n=z)} \circ \theta_m = 1_{(X_{n+m}=z)}$. $\square$

We are now ready for our second extension of the Markov property. Recall N is said to be a stopping time if $\{N = n\} \in \mathcal{F}_n$. As in Chapter 4, let

$$\mathcal{F}_N = \{A : A \cap \{N = n\} \in \mathcal{F}_n \text{ for all } n\}$$

be the information known at time N , and let

$$\theta_N \omega = \begin{cases} \theta_n \omega & \text{on } \{N = n\} \\ \Delta & \text{on } \{N = \infty\} \end{cases}$$

where Δ is an extra point that we add to Ω_0 . In the next result and its applications, we will explicitly restrict our attention to $\{N < \infty\}$, so the reader does not have to worry about the second part of the definition of θ_N .

Theorem 5.2.5 (Strong Markov property) *Suppose that for each n , $Y_n : \Omega_0 \rightarrow \mathbf{R}$ is measurable and $|Y_n| \leq M$ for all n . Then*

$$E_\mu(Y_N \circ \theta_N | \mathcal{F}_N) = E_{X_N} Y_N \text{ on } \{N < \infty\}$$

where the right-hand side is $\varphi(x, n) = E_x Y_n$ evaluated at $x = X_N$, $n = N$.

Proof Let $A \in \mathcal{F}_N$. Breaking things down according to the value of N .

$$E_\mu(Y_N \circ \theta_N; A \cap \{N < \infty\}) = \sum_{n=0}^{\infty} E_\mu(Y_n \circ \theta_n; A \cap \{N = n\})$$

Since $A \cap \{N = n\} \in \mathcal{F}_n$, using Theorem 5.2.3 now converts the right side into

$$\sum_{n=0}^{\infty} E_\mu(E_{X_n} Y_n; A \cap \{N = n\}) = E_\mu(E_{X_N} Y_N; A \cap \{N < \infty\}) \quad \square$$

Remark The reader should notice that the proof is trivial. All we do is break things down according to the value of N , replace N by n , apply the Markov property, and reverse the process. This is the standard technique for proving results about stopping times.

Our first application of the strong Markov property will be the key to developments in the next section. Let $T_y^0 = 0$, and for $k \geq 1$, let

$$T_y^k = \inf\{n > T_y^{k-1} : X_n = y\}$$

T_y^k is the time of the k th return to y . The reader should note that $T_y^1 > 0$, so any visit at time 0 does not count. We adopt this convention so that if we let $T_y = T_y^1$ and $\rho_{xy} = P_x(T_y < \infty)$, then:

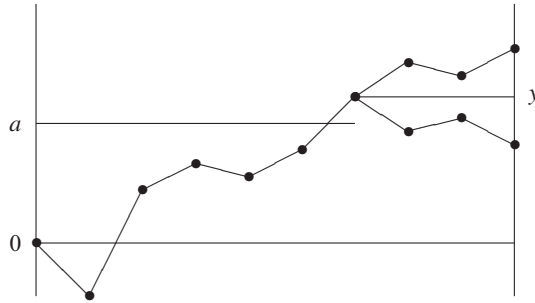


Figure 5.3 Proof by picture of the reflection principle.

Theorem 5.2.6 $P_x(T_y^k < \infty) = \rho_{xy}\rho_{yy}^{k-1}$.

Intuitively, in order to make k visits to y , we first have to go from x to y and then return $k - 1$ times to y .

Proof When $k = 1$, the result is trivial, so we suppose $k \geq 2$. Let $Y(\omega) = 1$ if $\omega_n = y$ for some $n \geq 1$, $Y(\omega) = 0$ otherwise. If $N = T_y^{k-1}$, then $Y \circ \theta_N = 1$ if $T_y^k < \infty$. The strong Markov property, Theorem 5.2.5, implies

$$E_x(Y \circ \theta_N | \mathcal{F}_N) = E_{X_N} Y \quad \text{on } \{N < \infty\}$$

On $\{N < \infty\}$, $X_N = y$, so the right-hand side is $P_y(T_y < \infty) = \rho_{yy}$, and it follows that

$$\begin{aligned} P_x(T_y^k < \infty) &= E_x(Y \circ \theta_N; N < \infty) \\ &= E_x(E_x(Y \circ \theta_N | \mathcal{F}_N); N < \infty) \\ &= E_x(\rho_{yy}; N < \infty) = \rho_{yy} P_x(T_y^{k-1} < \infty) \end{aligned}$$

The result now follows by induction. □

The next example illustrates the use of Theorem 5.2.5, and explains why we want to allow the Y that we apply to the shifted path to depend on n .

Theorem 5.2.7 (Reflection principle) Let $\xi_1, \xi_2, \dots$ be independent and identically distributed with a distribution that is symmetric about 0. Let $S_n = \xi_1 + \dots + \xi_n$. If $a > 0$, then

$$P\left(\sup_{m \leq n} S_m \geq a\right) \leq 2P(S_n \geq a)$$

We do the proof in two steps because that is how formulas like this are derived in practice. First, one computes intuitively and then figures out how to extract the desired formula from Theorem 5.2.5.

Proof in words. First note that if Z has a distribution that is symmetric about 0, then

$$P(Z \geq 0) \geq P(Z > 0) + \frac{1}{2}P(Z = 0) = \frac{1}{2}$$

If we let $N = \inf\{m \leq n : S_m > a\}$ (with $\inf \emptyset = \infty$), then on $\{N < \infty\}$, $S_n - S_N$ is independent of S_N and has $P(S_n - S_N \geq 0) \geq 1/2$. So

$$P(S_n > a) \geq \frac{1}{2} P(N \leq n)$$

Proof Let $Y_m(\omega) = 1$ if $m \leq n$ and $\omega_{n-m} \geq a$, $Y_m(\omega) = 0$ otherwise. The definition of Y_m is chosen so that $(Y_N \circ \theta_N)(\omega) = 1$ if $\omega_n \geq a$ (and hence $N \leq n$), and $= 0$ otherwise. The strong Markov property implies

$$E_0(Y_N \circ \theta_N | \mathcal{F}_N) = E_{S_N} Y_N \quad \text{on } \{N < \infty\} = \{N \leq n\}$$

To evaluate the right-hand side, we note that if $y > a$, then

$$E_y Y_m = P_y(S_{n-m} \geq a) \geq P_y(S_{n-m} \geq y) \geq 1/2$$

So integrating over $\{N \leq n\}$ and using the definition of conditional expectation gives

$$\frac{1}{2} P(N \leq n) \leq E_0(E_0(Y_N \circ \theta_N | \mathcal{F}_N); N \leq n) = E_0(Y_N \circ \theta_N; N \leq n)$$

since $\{N \leq n\} \in \mathcal{F}_N$. Recalling that $Y_N \circ \theta_N = 1_{\{S_n \geq a\}}$, the last quantity

$$= E_0(1_{\{S_n \geq a\}}; N \leq n) = P_0(S_n \geq a)$$

since $\{S_n > a\} \subset \{N \leq n\}$. □

Exercises

5.2.1 Let $A \in \sigma(X_0, \dots, X_n)$ and $B \in \sigma(X_n, X_{n+1}, \dots)$. Use the Markov property to show that for any initial distribution μ

$$P_\mu(A \cap B | X_n) = P_\mu(A | X_n) P_\mu(B | X_n)$$

In words, the past and future are conditionally independent given the present.

Hint: Write the left-hand side as $E_\mu(E_\mu(1_A 1_B | \mathcal{F}_n) | X_n)$.

5.2.2 Let X_n be a Markov chain. Use Lévy's 0-1 law to show that if

$$P\left(\bigcup_{m=n+1}^{\infty} \{X_m \in B_m\} \mid X_n\right) \geq \delta > 0 \quad \text{on } \{X_n \in A_n\}$$

then $P(\{X_n \in A_n \text{ i.o.}\} - \{X_n \in B_n \text{ i.o.}\}) = 0$.

5.2.3 A state a is called **absorbing** if $P_a(X_1 = a) = 1$. Let $D = \{X_n = a \text{ for some } n \geq 1\}$ and let $h(x) = P_x(D)$. Use the result of the previous exercise to conclude that $h(X_n) \rightarrow 0$ a.s. on D^c . Here a.s. means P_μ a.s. for any initial distribution μ .

We will suppose throughout the rest of the exercises that S is countable. Note that the times V_A and V_x of the first visit are inf over $n \geq 0$, while the hitting times T_A and T_x are inf over $n \geq 1$.

5.2.4 **First entrance decomposition.** Let $T_y = \inf\{n \geq 1 : X_n = y\}$. Show that

$$p^n(x, y) = \sum_{m=1}^n P_x(T_y = m) p^{n-m}(y, y)$$

- 5.2.5 Show that $\sum_{m=0}^n P_x(X_m = x) \geq \sum_{m=k}^{n+k} P_x(X_m = x)$.
- 5.2.6 Let $T_C = \inf\{n \geq 1 : X_n \in C\}$. Suppose that $S - C$ is finite and for each $x \in S - C$ $P_x(T_C < \infty) > 0$. Then there is an $N < \infty$ and $\epsilon > 0$ so that for all $y \in S - C$, $P_y(T_C > kN) \leq (1 - \epsilon)^k$.
- 5.2.7 **Exit distributions.** Let $V_C = \inf\{n \geq 0 : X_n \in C\}$ and let $h(x) = P_x(V_A < V_B)$. Suppose $A \cap B = \emptyset$, $S - (A \cup B)$ is finite, and $P_x(V_{A \cup B} < \infty) > 0$ for all $x \in S - (A \cup B)$. (i) Show that

$$(*) \quad h(x) = \sum_y p(x, y)h(y) \quad \text{for } x \notin A \cup B$$

- (ii) Show that if h satisfies (*), then $h(X(n \wedge V_{A \cup B}))$ is a martingale. (iii) Use this and Exercise 5.2.6 to conclude that $h(x) = P_x(V_A < V_B)$ is the only solution of (*) that is 1 on A and 0 on B .
- 5.2.8 Let X_n be a Markov chain with $S = \{0, 1, \dots, N\}$ and suppose that X_n is a martingale. Let $V_x = \min\{n \geq 0 : X_n = x\}$ and suppose $P_x(V_0 \wedge V_N < \infty) > 0$ for all x . Show $P_x(V_N < V_0) = x/N$.
- 5.2.9 **Wright-Fisher model.** Suppose $S = \{0, 1, \dots, N\}$ and consider

$$p(i, j) = \binom{N}{j} (i/N)^j (1 - i/N)^{N-j}$$

Show that this chain satisfies the hypotheses of Exercise 5.2.8.

- 5.2.10 In **brother-sister mating** described in Exercise 5.1.4, AA, AA and aa, aa are absorbing states. Show that the number of A 's in the pair is a martingale and use this to compute the probability of getting absorbed in AA, AA starting from each of the states.
- 5.2.11 **Exit Times.** Let $V_A = \inf\{n \geq 0 : X_n \in A\}$ and $g(x) = E_x V_A$. Suppose that $S - A$ is finite and for each $x \in S - A$, $P_x(V_A < \infty) > 0$. (i) Show that

$$(*) \quad g(x) = 1 + \sum_y p(x, y)g(y) \quad \text{for } x \notin A$$

- (ii) Show that if g satisfies (*), $g(X(n \wedge V_A)) + n \wedge V_A$ is a martingale. (iii) Use this to conclude that $g(x) = E_x \tau_A$ is the only solution of (*) that is 0 on A .
- 5.2.12 In this problem we imagine that you are bored and start flipping a coin to see how many tosses $N(HH)$ you need until you get two heads in a row. Let $\xi_0, \xi_1, \dots$ be i.i.d. $\in \{H, T\}$, taking each value with probability $1/2$, and let $X_n = (\xi_n, \xi_{n+1})$ be the Markov chain from Exercise 5.1.3. Use the result of the previous exercise to compute $g(x)$ = the expected time to reach state (H, H) starting from state x . The answer we want is

$$EN(HH) = 2 + (1/4)[g(H < H) + g(H, T) + g(T, H) + g(T, T)]$$

5.2.13 In this problem we consider the number of tosses $N(HHH)$ to get HHH , let X_n be the number of consecutive heads we have after n tosses. X_n is a Markov chain with transition probability

	0	1	2	3
0	1/2	1/2	0	0
1	1/2	0	1/2	0
2	1/2	0	0	1/2
3	0	0	0	1

Let T_3 be the time to reach state 3. Use Problem 5.2.11 to compute E_0T_3 .

5.3 Recurrence and Transience

Let $T_y^0 = 0$, and for $k \geq 1$, let

$$T_y^k = \inf\{n > T_y^{k-1} : X_n = y\}$$

and recall (5.2.6)

$$\rho_{xy} = P_x(T_y < \infty)$$

A state y is said to be **recurrent** if $\rho_{yy} = 1$ and **transient** if $\rho_{yy} < 1$. If y is recurrent, Theorem 5.2.6 implies $P_y(T_y^k < \infty) = 1$ for all k , so $P_y(X_n = y \text{ i.o.}) = 1$.

If y is transient and we let $N(y) = \sum_{n=1}^{\infty} 1_{(X_n=y)}$ be the number of visits to y at positive times, then

$$\begin{aligned} E_x N(y) &= \sum_{k=1}^{\infty} P_x(N(y) \geq k) = \sum_{k=1}^{\infty} P_x(T_y^k < \infty) \\ &= \sum_{k=1}^{\infty} \rho_{xy} \rho_{yy}^{k-1} = \frac{\rho_{xy}}{1 - \rho_{yy}} < \infty \end{aligned} \quad (5.3.1)$$

Combining the last computation with our result for recurrent states gives:

Theorem 5.3.1 y is recurrent if and only if $E_y N(y) = \infty$.

The next result shows that recurrence is contagious.

Theorem 5.3.2 If x is recurrent and $\rho_{xy} > 0$, then y is recurrent and $\rho_{yx} = 1$.

Proof We will first show $\rho_{yx} = 1$ by showing that if $\rho_{xy} > 0$ and $\rho_{yx} < 1$, then $\rho_{xx} < 1$. Let $K = \inf\{k : p^k(x, y) > 0\}$. There is a sequence $y_1, \dots, y_{K-1}$ so that

$$p(x, y_1)p(y_1, y_2) \cdots p(y_{K-1}, y) > 0$$

Since K is minimal, $y_i \neq x$ for $1 \leq i \leq K-1$. If $\rho_{yx} < 1$, we have

$$P_x(T_x = \infty) \geq p(x, y_1)p(y_1, y_2) \cdots p(y_{K-1}, y)(1 - \rho_{yx}) > 0$$

a contradiction. So $\rho_{yx} = 1$.

To prove that y is recurrent, observe that $\rho_{yx} > 0$ implies there is an L so that $p^L(y, x) > 0$. Now

$$p^{L+n+K}(y, y) \geq p^L(y, x)p^n(x, x)p^K(x, y)$$

Summing over n , we see

$$\sum_{n=1}^{\infty} p^{L+n+K}(y, y) \geq p^L(y, x)p^K(x, y) \sum_{n=1}^{\infty} p^n(x, x) = \infty$$

so Theorem 5.3.1 implies y is recurrent. $\square$

The next fact will help us identify recurrent states in examples. First we need two definitions. C is **closed** if $x \in C$ and $\rho_{xy} > 0$ implies $y \in C$. The name comes from the fact that if C is closed and $x \in C$, then $P_x(X_n \in C) = 1$ for all n . D is **irreducible** if $x, y \in D$ implies $\rho_{xy} > 0$.

Theorem 5.3.3 *Let C be a finite closed set. Then C contains a recurrent state. If C is irreducible then all states in C are recurrent.*

Proof In view of Theorem 5.3.2, it suffices to prove the first claim. Suppose it is false. Then for all $y \in C$, $\rho_{yy} < 1$ and $E_x N(y) = \rho_{xy}/(1 - \rho_{yy})$, but this is ridiculous since it implies

$$\infty > \sum_{y \in C} E_x N(y) = \sum_{y \in C} \sum_{n=1}^{\infty} p^n(x, y) = \sum_{n=1}^{\infty} \sum_{y \in C} p^n(x, y) = \sum_{n=1}^{\infty} 1$$

The first inequality follows from the fact that C is finite and the last equality from the fact that C is closed. $\square$

To illustrate the use of the last result consider:

Example 5.3.4 (A Seven-state chain) Consider the transition probability:

	1	2	3	4	5	6	7
1	.3	0	0	0	.7	0	0
2	.1	.2	.3	.4	0	0	0
3	0	0	.5	.5	0	0	0
4	0	0	0	.5	0	.5	0
5	.6	0	0	0	.4	0	0
6	0	0	0	.1	0	.1	.8
7	0	0	0	1	0	0	0

To identify the states that are recurrent and those that are transient, we begin by drawing a graph that will contain an arc from i to j if $p(i, j) > 0$ and $i \neq j$. We do not worry about drawing the self-loops corresponding to states with $p(i, i) > 0$, since such transitions cannot help the chain get somewhere new.

In the case under consideration, we draw arcs from $1 \rightarrow 5$, $2 \rightarrow 1$, $2 \rightarrow 3$, $2 \rightarrow 4$, $3 \rightarrow 4$, $4 \rightarrow 6$, $4 \rightarrow 7$, $5 \rightarrow 1$, $6 \rightarrow 7$, $7 \rightarrow 4$.

(i) $\rho_{21} > 0$ and $\rho_{12} = 0$, so 2 must be transient, or we would contradict Theorem 5.3.2. Similarly, $\rho_{34} > 0$ and $\rho_{43} = 0$ so 3 must be transient

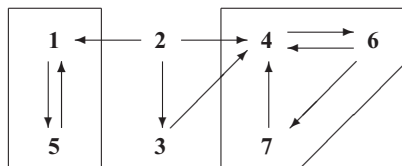


Figure 5.4 Graph for the seven state chain.

(ii) $\{1, 5\}$ and $\{4, 6, 7\}$ are irreducible closed sets, so Theorem 5.3.3 implies these states are recurrent.

The last reasoning can be used to identify transient and recurrent states when S is finite, since for $x \in S$ either: (i) there is a y with $\rho_{xy} > 0$ and $\rho_{yx} = 0$ and x must be transient, or (ii) $\rho_{xy} > 0$ implies $\rho_{yx} > 0$. In case (ii), Exercise 5.3.2 implies $C_x = \{y : \rho_{xy} > 0\}$ is an irreducible closed set. (If $y, z \in C_x$, then $\rho_{yz} \geq \rho_{yx}\rho_{xz} > 0$. If $\rho_{yw} > 0$, then $\rho_{xw} \geq \rho_{xy}\rho_{yw} > 0$, so $w \in C_x$.) So Theorem 5.3.3 implies x is recurrent.

Example 5.3.4 motivates the following:

Theorem 5.3.5 (Decomposition theorem) Let $R = \{x : \rho_{xx} = 1\}$ be the recurrent states of a Markov chain. R can be written as $\cup_i R_i$, where each R_i is closed and irreducible.

Remark This result shows that for the study of recurrent states we can, without loss of generality, consider a single irreducible closed set.

Proof If $x \in R$, let $C_x = \{y : \rho_{xy} > 0\}$. By Theorem 5.3.2, $C_x \subset R$, and if $y \in C_x$, then $\rho_{yx} > 0$. From this it follows easily that either $C_x \cap C_y = \emptyset$ or $C_x = C_y$. To prove the last claim, suppose $C_x \cap C_y \neq \emptyset$. If $z \in C_x \cap C_y$, then $\rho_{xy} \geq \rho_{xz}\rho_{zy} > 0$, so if $w \in C_y$, we have $\rho_{xw} \geq \rho_{xy}\rho_{yw} > 0$ and it follows that $C_x \supset C_y$. Interchanging the roles of x and y gives $C_y \supset C_x$, and we have proved our claim. If we let R_i be a listing of the sets that appear as some C_x , we have the desired decomposition. $\square$

The rest of this section is devoted to examples. Specifically, we concentrate on the question: How do we tell whether a state is recurrent or transient? Reasoning based on Theorem 5.3.2 works occasionally when S is infinite.

Example 5.3.6 (Branching process) If the probability of no children is positive, then $\rho_{k0} > 0$ and $\rho_{0k} = 0$ for $k \geq 1$, so Theorem 5.3.3 implies all states $k \geq 1$ are transient. The state 0 has $p(0, 0) = 1$ and is recurrent. It is called an **absorbing state** to reflect the fact that once the chain enters 0, it remains there for all time.

If S is infinite and irreducible, all that Theorem 5.3.2 tells us is that either all the states are recurrent or all are transient, and we are left to figure out which case occurs.

Example 5.3.7 (M/G/1 queue) Let $\mu = \sum k a_k$ be the mean number of customers that arrive during one service time. We will now show that if $\mu > 1$, the chain is transient (i.e., all states are), but if $\mu \leq 1$, it is recurrent. For the case $\mu > 1$, we observe that if $\xi_1, \xi_2, \dots$ are i.i.d. with $P(\xi_m = j) = a_{j+1}$ for $j \geq -1$ and $S_n = \xi_1 + \dots + \xi_n$, then $X_0 + S_n$ and X_n behave the same until time $N = \inf\{n : X_0 + S_n = 0\}$. When $\mu > 1$, $E\xi_m = \mu - 1 > 0$,

so $S_n \rightarrow \infty$ a.s., and $\inf S_n > -\infty$ a.s. It follows from the last observation that if x is large, $P_x(N < \infty) < 1$, and the chain is transient.

To deal with the case $\mu \leq 1$, we observe that it follows from arguments in the last paragraph that $X_{n \wedge N}$ is a supermartingale. Let $T = \inf\{n : X_n \geq M\}$. Since $X_{n \wedge N}$ is a nonnegative supermartingale, using Theorem 4.8.4 at time $\tau = T \wedge N$, and observing $X_\tau \geq M$ on $\{T < N\}$, $X_\tau = 0$ on $\{N < T\}$ gives

$$x \geq M P_x(T < N)$$

Letting $M \rightarrow \infty$ shows $P_x(N < \infty) = 1$, so the chain is recurrent.

Remark There is another way of seeing that the $M/G/1$ queue is transient when $\mu > 1$. If we consider the customers that arrive during a person's service time to be her children, then we get a branching process. Results in Section 5.3 imply that when $\mu \leq 1$, the branching process dies out with probability one (i.e., the queue becomes empty), so the chain is recurrent. When $\mu > 1$, Theorem 4.3.12 implies $P_x(T_0 < \infty) = \rho^x$, where ρ is the unique fixed point $\in (0, 1)$ of the function $\varphi(\theta) = \sum_{k=0}^{\infty} a_k \theta^k$.

The next result encapsulates the techniques we used for birth and death chains and the $M/G/1$ queue.

Theorem 5.3.8 *Suppose S is irreducible, and $\varphi \geq 0$ with $E_x \varphi(X_1) \leq \varphi(x)$ for $x \notin F$, a finite set, and $\varphi(x) \rightarrow \infty$ as $x \rightarrow \infty$, i.e., $\{x : \varphi(x) \leq M\}$ is finite for any $M < \infty$, then the chain is recurrent.*

Proof Let $\tau = \inf\{n > 0 : X_n \in F\}$. Our assumptions imply that $Y_n = \varphi(X_{n \wedge \tau})$ is a supermartingale. Let $T_M = \inf\{n > 0 : X_n \in F \text{ or } \varphi(X_n) > M\}$. Since $\{x : \varphi(x) \leq M\}$ is finite and the chain is irreducible, $T_M < \infty$ a.s. Using Theorem 4.8.4 now, we see that

$$\varphi(x) \geq E_x \varphi(X_{T_M}) \geq M P_x(T_M < \tau)$$

since $\varphi(X_{T_M}) \geq M$ when $T_M < \tau$. Letting $M \rightarrow \infty$, we see that $P_x(\tau < \infty) = 1$ for all $x \notin F$. So $P_y(X_n \in F \text{ i.o.}) = 1$ for all $y \in S$, and since F is finite, $P_y(X_n = z \text{ i.o.}) = 1$ for some $z \in F$. $\square$

Example 5.3.9 (Birth and death chains on $\{0, 1, 2, \dots\}$) Let

$$p(i, i+1) = p_i \quad p(i, i-1) = q_i \quad p(i, i) = r_i$$

where $q_0 = 0$. Let $N = \inf\{n : X_n = 0\}$. To analyze this example, we are going to define a function φ so that $\varphi(X_{N \wedge n})$ is a martingale. We start by setting $\varphi(0) = 0$ and $\varphi(1) = 1$. For the martingale property to hold when $X_n = k \geq 1$, we must have

$$\varphi(k) = p_k \varphi(k+1) + r_k \varphi(k) + q_k \varphi(k-1)$$

Using $r_k = 1 - (p_k + q_k)$, we can rewrite the last equation as

$$q_k(\varphi(k) - \varphi(k-1)) = p_k(\varphi(k+1) - \varphi(k))$$

$$\text{or } \varphi(k+1) - \varphi(k) = \frac{q_k}{p_k}(\varphi(k) - \varphi(k-1))$$

Here and in what follows, we suppose that $p_k, q_k > 0$ for $k \geq 1$. Otherwise, the chain is not irreducible. Since $\varphi(1) - \varphi(0) = 1$, iterating the last result gives

$$\begin{aligned}\varphi(m+1) - \varphi(m) &= \prod_{j=1}^m \frac{q_j}{p_j} \quad \text{for } m \geq 1 \\ \varphi(n) &= \sum_{m=0}^{n-1} \prod_{j=1}^m \frac{q_j}{p_j} \quad \text{for } n \geq 1\end{aligned}$$

if we interpret the product as 1 when $m = 0$. Let $T_c = \inf\{n \geq 1 : X_n = c\}$. Now I claim that:

Theorem 5.3.10 *If $a < x < b$, then*

$$P_x(T_a < T_b) = \frac{\varphi(b) - \varphi(x)}{\varphi(b) - \varphi(a)} \quad P_x(T_b < T_a) = \frac{\varphi(x) - \varphi(a)}{\varphi(b) - \varphi(a)}$$

Proof If we let $T = T_a \wedge T_b$, then $\varphi(X_{n \wedge T})$ is a bounded martingale, so $\varphi(x) = E_x \varphi(X_T)$ by Theorem 4.8.2. Since $X_T \in \{a, b\}$ a.s.,

$$\varphi(x) = \varphi(a)P_x(T_a < T_b) + \varphi(b)[1 - P_x(T_a < T_b)]$$

and solving gives the indicated formula. $\square$

Remark The answer and the proof should remind the reader of our results for symmetric and asymmetric simple random walk, which are special cases of this.

Letting $a = 0$ and $b = M$ in Theorem 5.3.10 gives

$$P_x(T_0 > T_M) = \varphi(x)/\varphi(M)$$

Letting $M \rightarrow \infty$ and observing that $T_M \geq M - x$, P_x a.s., we have proved:

Theorem 5.3.11 *0 is recurrent if and only if $\varphi(M) \rightarrow \infty$ as $M \rightarrow \infty$, i.e.,*

$$\varphi(\infty) \equiv \sum_{m=0}^{\infty} \prod_{j=1}^m \frac{q_j}{p_j} = \infty$$

If $\varphi(\infty) < \infty$, then $P_x(T_0 = \infty) = \varphi(x)/\varphi(\infty)$.

Exercises

5.3.1 Suppose y is recurrent and for $k \geq 0$, let $R_k = T_y^k$ be the time of the k th return to y , and for $k \geq 1$ let $r_k = R_k - R_{k-1}$ be the k th interarrival time. Use the strong Markov property to conclude that under P_y , the vectors $v_k = (r_k, X_{R_{k-1}}, \dots, X_{R_k-1})$, $k \geq 1$ are i.i.d.

5.3.2 Use the strong Markov property to show that $\rho_{xz} \geq \rho_{xy}\rho_{yz}$.

5.3.3 Show that in the Ehrenfest chain (Example 5.1.3) all states are recurrent.

- 5.3.4 To probe the boundary between recurrence and transience for birth and death chains, suppose $p_j = 1/2 + \epsilon_j$, where $\epsilon_j \sim Cj^{-\alpha}$ as $j \rightarrow \infty$, and $q_j = 1 - p_j$. Show that (i) if $\alpha > 1$ 0 is recurrent, (ii) if $\alpha < 1$ 0 is transient, (iii) if $\alpha = 1$, then 0 is transient if $C > 1/4$ and recurrent if $C < 1/4$.
- 5.3.5 Show that if we replace “ $\varphi(x) \rightarrow \infty$ ” by “ $\varphi(x) \rightarrow 0$ ” in the Theorem 5.3.8 and assume that $\varphi(x) > 0$ for $x \in F$, then we can conclude that the chain is transient.
- 5.3.6 Let X_n be a birth and death chain with $p_j - 1/2 \sim C/j$ as $j \rightarrow \infty$ and $q_j = 1 - p_j$. (i) Show that if we take $C < 1/4$, then we can pick $\alpha > 0$ so that $\varphi(x) = x^\alpha$ satisfies the hypotheses of Theorem 5.3.8. (ii) Show that when $C > 1/4$, we can take $\alpha < 0$ and apply Exercise 5.3.5.
- 5.3.7 f is said to be **superharmonic** if $f(x) \geq \sum_y p(x, y)f(y)$, or equivalently $f(X_n)$ is a supermartingale. Suppose p is irreducible. Show that p is recurrent if and only if every nonnegative superharmonic function is constant.
- 5.3.8 **M/M/ ∞ queue.** Consider a telephone system with an infinite number of lines. Let X_n = the number of lines in use at time n , and suppose

$$X_{n+1} = \sum_{m=1}^{X_n} \xi_{n,m} + Y_{n+1}$$

where the $\xi_{n,m}$ are i.i.d. with $P(\xi_{n,m} = 1) = p$ and $P(\xi_{n,m} = 0) = 1 - p$, and Y_n is an independent i.i.d. sequence of Poisson mean λ r.v.'s. In words, for each conversation we flip a coin with probability p of heads to see if it continues for another minute. Meanwhile, a Poisson mean λ number of conversations start between time n and $n+1$. Use Theorem 5.3.8 with $\varphi(x) = x$ to show that the chain is recurrent for any $p < 1$.

5.4 Recurrence of Random Walks Stararred Section

Throughout this section, S_n will be a random walk, i.e., $S_n = X_1 + \cdots + X_n$, where $X_1, X_2, \dots$ are i.i.d. The number $x \in \mathbf{R}^d$ is said to be a **recurrent value** for the random walk S_n if for every $\epsilon > 0$, $P(\|S_n - x\| < \epsilon \text{ i.o.}) = 1$. Here $\|x\| = \sup |x_i|$. The reader will see the reason for this choice of norm in the proof of Lemma 5.4.6. The Hewitt-Savage 0-1 law, Theorem 2.5.4, implies that if the last probability is < 1 , it is 0. Our first result shows that to know the set of recurrent values, it is enough to check $x = 0$. A number x is said to be a **possible value** of the random walk if for any $\epsilon > 0$, there is an n so that $P(\|S_n - x\| < \epsilon) > 0$.

Theorem 5.4.1 *The set $\mathcal{V}$ of recurrent values is either $\emptyset$ or a closed subgroup of $\mathbf{R}^d$. In the second case, $\mathcal{V} = \mathcal{U}$, the set of possible values.*

Proof Suppose $\mathcal{V} \neq \emptyset$. It is clear that $\mathcal{V}^c$ is open, so $\mathcal{V}$ is closed. To prove that $\mathcal{V}$ is a group, we will first show that

(*) if $x \in \mathcal{U}$ and $y \in \mathcal{V}$, then $y - x \in \mathcal{V}$.

This statement has been formulated so that once it is established, the result follows easily. Let

$$p_{\delta,m}(z) = P(\|S_n - z\| \geq \delta \text{ for all } n \geq m)$$

If $y - x \notin \mathcal{V}$, there is an $\epsilon > 0$ and $m \geq 1$ so that $p_{2\epsilon,m}(y - x) > 0$. Since $x \in \mathcal{U}$, there is a k so that $P(\|S_k - x\| < \epsilon) > 0$. Since

$$P(\|S_n - S_k - (y - x)\| \geq 2\epsilon \text{ for all } n \geq k + m) = p_{2\epsilon,m}(y - x)$$

and is independent of $\{\|S_k - x\| < \epsilon\}$, it follows that

$$p_{\epsilon,m+k}(y) \geq P(\|S_k - x\| < \epsilon) p_{2\epsilon,m}(y - x) > 0$$

contradicting $y \in \mathcal{V}$, so $y - x \in \mathcal{V}$.

To conclude $\mathcal{V}$ is a group when $\mathcal{V} \neq \emptyset$, let $q, r \in \mathcal{V}$, and observe: (i) taking $x = y = r$ in (*) shows $0 \in \mathcal{V}$, (ii) taking $x = r$, $y = 0$ shows $-r \in \mathcal{V}$, and (iii) taking $x = -r$, $y = q$ shows $q + r \in \mathcal{V}$. To prove that $\mathcal{V} = \mathcal{U}$ now, observe that if $u \in \mathcal{U}$ taking $x = u$, $y = 0$ shows $-u \in \mathcal{V}$ and since $\mathcal{V}$ is a group, it follows that $u \in \mathcal{V}$. $\square$

If $\mathcal{V} = \emptyset$, the random walk is said to be **transient**, otherwise it is called **recurrent**. Before plunging into the technicalities needed to treat a general random walk, we begin by analyzing the special case Polya considered in 1921. Legend has it that Polya thought of this problem while wandering around in a park near Zürich when he noticed that he kept encountering the same young couple. History does not record what the young couple thought.

Example 5.4.2 (Simple random walk on $\mathbb{Z}^d$)

$$P(X_i = e_j) = P(X_i = -e_j) = 1/2d$$

for each of the d unit vectors e_j . To analyze this case, we begin with a result that is valid for any random walk. Let $\tau_0 = 0$ and $\tau_n = \inf\{m > \tau_{n-1} : S_m = 0\}$ be the time of the n th return to 0. From the strong Markov property, it follows that

$$P(\tau_n < \infty) = P(\tau_1 < \infty)^n$$

a fact that leads easily to:

Theorem 5.4.3 *For any random walk, the following are equivalent:*

(i) $P(\tau_1 < \infty) = 1$, (ii) $P(S_m = 0 \text{ i.o.}) = 1$, and (iii) $\sum_{m=0}^{\infty} P(S_m = 0) = \infty$.

Proof If $P(\tau_1 < \infty) = 1$, then $P(\tau_n < \infty) = 1$ for all n and $P(S_m = 0 \text{ i.o.}) = 1$. Let

$$V = \sum_{m=0}^{\infty} 1_{(S_m=0)} = \sum_{n=0}^{\infty} 1_{(\tau_n < \infty)}$$

be the number of visits to 0, counting the visit at time 0. Taking expected value and using Fubini's theorem to put the expected value inside the sum:

$$\begin{aligned} EV &= \sum_{m=0}^{\infty} P(S_m = 0) = \sum_{n=0}^{\infty} P(\tau_n < \infty) \\ &= \sum_{n=0}^{\infty} P(\tau_1 < \infty)^n = \frac{1}{1 - P(\tau_1 < \infty)} \end{aligned}$$

The second equality shows (ii) implies (iii), and in combination with the last two shows that if (i) is false, then (iii) is false (i.e., (iii) implies (i)). $\square$

Theorem 5.4.4 *Simple random walk is recurrent in $d \leq 2$ and transient in $d \geq 3$.*

To steal a joke from Kakutani (U.C.L.A. colloquium talk): “A drunk man will eventually find his way home but a drunk bird may get lost forever.”

Proof Let $\rho_d(m) = P(S_m = 0)$. $\rho_d(m)$ is 0 if m is odd. From Theorem 3.1.3, we get $\rho_1(2n) \sim (\pi n)^{-1/2}$ as $n \rightarrow \infty$. This and Theorem 5.4.3 give the result in one dimension. Our next step is

Simple random walk is recurrent in two dimensions. Note that in order for $S_{2n} = 0$, we must for some $0 \leq m \leq n$ have m up steps, m down steps, $n - m$ to the left, and $n - m$ to the right, so

$$\begin{aligned} \rho_2(2n) &= 4^{-2n} \sum_{m=0}^n \frac{2n!}{m! m! (n-m)! (n-m)!} \\ &= 4^{-2n} \binom{2n}{n} \sum_{m=0}^n \binom{n}{m} \binom{n}{n-m} = 4^{-2n} \binom{2n}{n}^2 = \rho_1(2n)^2 \end{aligned}$$

To see the next to last equality, consider choosing n students from a class with n boys and n girls and observe that for some $0 \leq m \leq n$ you must choose m boys and $n - m$ girls. Using the asymptotic formula $\rho_1(2n) \sim (\pi n)^{-1/2}$, we get $\rho_2(2n) \sim (\pi n)^{-1}$. Since $\sum n^{-1} = \infty$, the result follows from Theorem 5.4.3.

Remark For a direct proof of $\rho_2(2n) = \rho_1(2n)^2$, note that if T_n^1 and T_n^2 are independent, one-dimensional random walks, then T_n jumps from x to $x + (1, 1)$, $x + (1, -1)$, $x + (-1, 1)$, and $x + (-1, -1)$ with equal probability, so rotating T_n by 45 degrees and dividing by $\sqrt{2}$ gives S_n .

Simple random walk is transient in three dimensions. Intuitively, this holds, since the probability of being back at 0 after $2n$ steps is $\sim cn^{-3/2}$ and this is summable. We will not compute the probability exactly but will get an upper bound of the right order of magnitude. Again, since the number of steps in the directions $\pm e_i$ must be equal for $i = 1, 2, 3$

$$\begin{aligned} \rho_3(2n) &= 6^{-2n} \sum_{j,k} \frac{(2n)!}{(j! k! (n-j-k)!)^2} \\ &= 2^{-2n} \binom{2n}{n} \sum_{j,k} \left(3^{-n} \frac{n!}{j! k! (n-j-k)!} \right)^2 \\ &\leq 2^{-2n} \binom{2n}{n} \max_{j,k} 3^{-n} \frac{n!}{j! k! (n-j-k)!} \end{aligned}$$

where in the last inequality we have used the fact that if $a_{j,k}$ are ≥ 0 and sum to 1, then $\sum_{j,k} a_{j,k}^2 \leq \max_{j,k} a_{j,k}$. Our last step is to show

$$\max_{j,k} 3^{-n} \frac{n!}{j! k! (n-j-k)!} \leq Cn^{-1}$$

To do this, we note that (a) if any of the numbers j , k , or $n - j - k$ is $< [n/3]$ increasing the smallest number and decreasing the largest number decreases the denominator (since $x(1-x)$ is maximized at $1/2$), so the maximum occurs when all three numbers are as close as possible to $n/3$; (b) Stirling's formula implies

$$\frac{n!}{j!k!(n-j-k)!} \sim \frac{n^n}{j^j k^k (n-j-k)^{n-j-k}} \cdot \sqrt{\frac{n}{jk(n-j-k)}} \cdot \frac{1}{2\pi}$$

Taking j and k within 1 of $n/3$, the first term on the right is $\leq C3^n$, and the desired result follows.

Simple random walk is transient in $d > 3$. Let $T_n = (S_n^1, S_n^2, S_n^3)$, $N(0) = 0$ and $N(n) = \inf\{m > N(n-1) : T_m \neq T_{N(n-1)}\}$. It is easy to see that $T_{N(n)}$ is a three-dimensional simple random walk. Since $T_{N(n)}$ returns infinitely often to 0 with probability 0 and the first three coordinates are constant in between the $N(n)$, S_n is transient. $\square$

The rest of this section is devoted to proving the following facts about random walks:

- S_n is recurrent in $d = 1$ if $S_n/n \rightarrow 0$ in probability.
- S_n is recurrent in $d = 2$ if $S_n/n^{1/2} \Rightarrow$ a nondegenerate normal distribution.
- S_n is transient in $d \geq 3$ if it is “truly three dimensional.”

To prove the last result, we will give a necessary and sufficient condition for recurrence.

The first step in deriving these results is to generalize Theorem 5.4.3.

Lemma 5.4.5 *If $\sum_{n=1}^{\infty} P(\|S_n\| < \epsilon) < \infty$, then $P(\|S_n\| < \epsilon \text{ i.o.}) = 0$.
If $\sum_{n=1}^{\infty} P(\|S_n\| < \epsilon) = \infty$, then $P(\|S_n\| < 2\epsilon \text{ i.o.}) = 1$.*

Proof The first conclusion follows from the Borel-Cantelli lemma. To prove the second, let $F = \{\|S_n\| < \epsilon \text{ i.o.}\}^c$. Breaking things down according to the last time $\|S_n\| < \epsilon$,

$$\begin{aligned} P(F) &= \sum_{m=0}^{\infty} P(\|S_m\| < \epsilon, \|S_n\| \geq \epsilon \text{ for all } n \geq m+1) \\ &\geq \sum_{m=0}^{\infty} P(\|S_m\| < \epsilon, \|S_n - S_m\| \geq 2\epsilon \text{ for all } n \geq m+1) \\ &= \sum_{m=0}^{\infty} P(\|S_m\| < \epsilon) \rho_{2\epsilon, 1} \end{aligned}$$

where $\rho_{\delta, k} = P(\|S_n\| \geq \delta \text{ for all } n \geq k)$. Since $P(F) \leq 1$, and

$$\sum_{m=0}^{\infty} P(\|S_m\| < \epsilon) = \infty$$

it follows that $\rho_{2\epsilon, 1} = 0$. To extend this conclusion to $\rho_{2\epsilon, k}$ with $k \geq 2$, let

$$A_m = \{\|S_m\| < \epsilon, \|S_n\| \geq \epsilon \text{ for all } n \geq m+k\}$$

Since any ω can be in at most k of the A_m , repeating the argument gives

$$k \geq \sum_{m=0}^{\infty} P(A_m) \geq \sum_{m=0}^{\infty} P(\|S_m\| < \epsilon) \rho_{2\epsilon, k}$$

So $\rho_{2\epsilon, k} = P(\|S_n\| \geq 2\epsilon \text{ for all } j \geq k) = 0$, and since k is arbitrary, the desired conclusion follows. $\square$

Our second step is to show that the convergence or divergence of the sums in Lemma 5.4.5 is independent of ϵ . The previous proof works for any norm. For the next one, we need $\|x\| = \sup_i |x_i|$.

Lemma 5.4.6 *Let m be an integer ≥ 2 .*

$$\sum_{n=0}^{\infty} P(\|S_n\| < m\epsilon) \leq (2m)^d \sum_{n=0}^{\infty} P(\|S_n\| < \epsilon)$$

Proof We begin by observing

$$\sum_{n=0}^{\infty} P(\|S_n\| < m\epsilon) \leq \sum_{n=0}^{\infty} \sum_k P(S_n \in k\epsilon + [0, \epsilon)^d)$$

where the inner sum is over $k \in \{-m, \dots, m-1\}^d$. If we let

$$T_k = \inf\{\ell \geq 0 : S_\ell \in k\epsilon + [0, \epsilon)^d\}$$

then breaking things down according to the value of T_k and using Fubini's theorem gives

$$\begin{aligned} \sum_{n=0}^{\infty} P(S_n \in k\epsilon + [0, \epsilon)^d) &= \sum_{n=0}^{\infty} \sum_{\ell=0}^n P(S_n \in k\epsilon + [0, \epsilon)^d, T_k = \ell) \\ &\leq \sum_{\ell=0}^{\infty} \sum_{n=\ell}^{\infty} P(\|S_n - S_\ell\| < \epsilon, T_k = \ell) \end{aligned}$$

Since $\{T_k = \ell\}$ and $\{\|S_n - S_\ell\| < \epsilon\}$ are independent, the last sum

$$= \sum_{m=0}^{\infty} P(T_k = m) \sum_{j=0}^{\infty} P(\|S_j\| < \epsilon) \leq \sum_{j=0}^{\infty} P(\|S_j\| < \epsilon)$$

Since there are $(2m)^d$ values of k in $\{-m, \dots, m-1\}^d$, the proof is complete. $\square$

Combining Lemmas 5.4.5 and 5.4.6 gives:

Theorem 5.4.7 *The convergence (resp. divergence) of $\sum_n P(\|S_n\| < \epsilon)$ for a single value of $\epsilon > 0$ is sufficient for transience (resp. recurrence).*

In $d = 1$, if $EX_i = \mu \neq 0$, then the strong law of large numbers implies $S_n/n \rightarrow \mu$ so $|S_n| \rightarrow \infty$ and S_n is transient. As a converse, we have:

Theorem 5.4.8 (Chung-Fuchs theorem) *Suppose $d = 1$. If the weak law of large numbers holds in the form $S_n/n \rightarrow 0$ in probability, then S_n is recurrent.*

Proof Let $u_n(x) = P(|S_n| < x)$ for $x > 0$. Lemma 5.4.6 implies

$$\sum_{n=0}^{\infty} u_n(1) \geq \frac{1}{2m} \sum_{n=0}^{\infty} u_n(m) \geq \frac{1}{2m} \sum_{n=0}^{Am} u_n(n/A)$$

for any $A < \infty$, since $u_n(x) \geq 0$ and is increasing in x . By hypothesis $u_n(n/A) \rightarrow 1$, so letting $m \rightarrow \infty$ and noticing the right-hand side is $A/2$ times the average of the first Am terms

$$\sum_{n=0}^{\infty} u_n(1) \geq A/2$$

Since A is arbitrary, the sum must be ∞ , and the desired conclusion follows from Theorem 5.4.7. $\square$

Theorem 5.4.9 *If S_n is a random walk in $\mathbf{R}^2$ and $S_n/n^{1/2} \Rightarrow$ a nondegenerate normal distribution, then S_n is recurrent.*

Remark The conclusion is also true if the limit is degenerate, but in that case the random walk is essentially one- (or zero-) dimensional, and the result follows from the Chung-Fuchs theorem.

Proof Let $u(n, m) = P(\|S_n\| < m)$. Lemma 5.4.6 implies

$$\sum_{n=0}^{\infty} u(n, 1) \geq (4m^2)^{-1} \sum_{n=0}^{\infty} u(n, m)$$

If $m/\sqrt{n} \rightarrow c$, then

$$u(n, m) \rightarrow \int_{[-c, c]^2} n(x) dx$$

where $n(x)$ is the density of the limiting normal distribution. If we use $\rho(c)$ to denote the right-hand side and let $n = [\theta m^2]$, it follows that $u([\theta m^2], m) \rightarrow \rho(\theta^{-1/2})$. If we write

$$m^{-2} \sum_{n=0}^{\infty} u(n, m) = \int_0^{\infty} u([\theta m^2], m) d\theta$$

let $m \rightarrow \infty$, and use Fatou's lemma, we get

$$\liminf_{m \rightarrow \infty} (4m^2)^{-1} \sum_{n=0}^{\infty} u(n, m) \geq 4^{-1} \int_0^{\infty} \rho(\theta^{-1/2}) d\theta$$

Since the normal density is positive and continuous at 0

$$\rho(c) = \int_{[-c, c]^2} n(x) dx \sim n(0)(2c)^2$$

as $c \rightarrow 0$. So $\rho(\theta^{-1/2}) \sim 4n(0)/\theta$ as $\theta \rightarrow \infty$, the integral diverges, and backtracking to the first inequality in the proof it follows that $\sum_{n=0}^{\infty} u(n, 1) = \infty$, proving the result. $\square$

We come now to the promised necessary and sufficient condition for recurrence. Here $\phi = E \exp(it \cdot X_j)$ is the ch.f. of one step of the random walk.

Theorem 5.4.10 *Let $\delta > 0$. S_n is recurrent if and only if*

$$\int_{(-\delta, \delta)^d} \operatorname{Re} \frac{1}{1 - \varphi(y)} dy = \infty$$

We will prove a weaker result:

Theorem 5.4.11 *Let $\delta > 0$. S_n is recurrent if and only if*

$$\sup_{r < 1} \int_{(-\delta, \delta)^d} \operatorname{Re} \frac{1}{1 - r\varphi(y)} dy = \infty$$

Remark Half of the work needed to get the first result from the second is trivial.

$$0 \leq \operatorname{Re} \frac{1}{1 - r\varphi(y)} \rightarrow \operatorname{Re} \frac{1}{1 - \varphi(y)} \quad \text{as } r \rightarrow 1$$

so Fatou's lemma shows that if the integral is infinite, the walk is recurrent. The other direction is rather difficult: the second result is in Chung and Fuchs (1951), but a proof of the first result had to wait for Ornstein (1969) and Stone (1969) to solve the problem independently. Their proofs use a trick to reduce to the case where the increments have a density and then a second trick to deal with that case, so we will not give the details here. The reader can consult either of the sources cited or Port and Stone (1969), where the result is demonstrated for random walks on Abelian groups.

Proof The first ingredient in the solution is the:

Lemma 5.4.12 (Parseval relation) *Let μ and ν be probability measures on $\mathbf{R}^d$ with ch.f.'s φ and ψ .*

$$\int \psi(t) \mu(dt) = \int \varphi(x) \nu(dx)$$

Proof Since $e^{it \cdot x}$ is bounded, Fubini's theorem implies

$$\int \psi(t) \mu(dt) = \int \int e^{itx} \nu(dx) \mu(dt) = \int \int e^{itx} \mu(dt) \nu(dx) = \int \varphi(x) \nu(dx) \quad \square$$

Our second ingredient is a little calculus.

Lemma 5.4.13 *If $|x| \leq \pi/3$, then $1 - \cos x \geq x^2/4$.*

Proof It suffices to prove the result for $x > 0$. If $z \leq \pi/3$, then $\cos z \geq 1/2$,

$$\begin{aligned} \sin y &= \int_0^y \cos z dz \geq \frac{y}{2} \\ 1 - \cos x &= \int_0^x \sin y dy \geq \int_0^x \frac{y}{2} dy = \frac{x^2}{4} \end{aligned}$$

which proves the desired result. $\square$

From Example 3.3.7, we see that the density

$$\frac{\delta - |x|}{\delta^2} \quad \text{when } |x| \leq \delta, \quad 0 \quad \text{otherwise}$$

has ch.f. $2(1 - \cos \delta t)/(\delta t)^2$. Let μ_n denote the distribution of S_n . Using Lemma 5.4.13 (note $\pi/3 \geq 1$) and then Lemma 5.4.12, we have

$$\begin{aligned} P(\|S_n\| < 1/\delta) &\leq 4^d \int \prod_{i=1}^d \frac{1 - \cos(\delta t_i)}{(\delta t_i)^2} \mu_n(dt) \\ &= 2^d \int_{(-\delta, \delta)^d} \prod_{i=1}^d \frac{\delta - |x_i|}{\delta^2} \varphi^n(x) dx \end{aligned}$$

Our next step is to sum from 0 to ∞ . To be able to interchange the sum and the integral, we first multiply by r^n , where $r < 1$.

$$\sum_{n=0}^{\infty} r^n P(\|S_n\| < 1/\delta) \leq 2^d \int_{(-\delta, \delta)^d} \prod_{i=1}^d \frac{\delta - |x_i|}{\delta^2} \frac{1}{1 - r\varphi(x)} dx$$

Symmetry dictates that the integral on the right is real, so we can take the real part without affecting its value. Letting $r \uparrow 1$ and using $(\delta - |x|)/\delta \leq 1$

$$\sum_{n=0}^{\infty} P(\|S_n\| < 1/\delta) \leq \left(\frac{2}{\delta}\right)^d \sup_{r < 1} \int_{(-\delta, \delta)^d} \operatorname{Re} \frac{1}{1 - r\varphi(x)} dx$$

and using Theorem 5.4.7 gives half of Theorem 5.4.11.

To prove the other direction, we begin by noting that Example 3.3.15 shows that the density $(1 - \cos(x/\delta))/\pi x^2/\delta$ has ch.f. $1 - |\delta t|$ when $|t| \leq 1/\delta$, 0 otherwise. Using $1 \geq \prod_{i=1}^d (1 - |\delta x_i|)$ and then Lemma 5.4.12,

$$\begin{aligned} P(\|S_n\| < 1/\delta) &\geq \int_{(-1/\delta, 1/\delta)^d} \prod_{i=1}^d (1 - |\delta x_i|) \mu_n(dx) \\ &= \int \prod_{i=1}^d \frac{1 - \cos(t_i/\delta)}{\pi t_i^2/\delta} \varphi^n(t) dt \end{aligned}$$

Multiplying by r^n and summing gives

$$\sum_{n=0}^{\infty} r^n P(\|S_n\| < 1/\delta) \geq \int \prod_{i=1}^d \frac{1 - \cos(t_i/\delta)}{\pi t_i^2/\delta} \frac{1}{1 - r\varphi(t)} dt$$

The last integral is real, so its value is unaffected if we integrate only the real part of the integrand. If we do this and apply Lemma 5.4.13, we get

$$\sum_{n=0}^{\infty} r^n P(\|S_n\| < 1/\delta) \geq (4\pi\delta)^{-d} \int_{(-\delta, \delta)^d} \operatorname{Re} \frac{1}{1 - r\varphi(t)} dt$$

Letting $r \uparrow 1$ and using Theorem 5.4.7 now completes the proof of Theorem 5.4.11. $\square$

We will now consider some examples. Our goal in $d = 1$ and $d = 2$ is to convince you that the conditions in Theorems 5.4.8 and 5.4.9 are close to the best possible.

$d = 1$. Consider the symmetric stable laws that have ch.f. $\varphi(t) = \exp(-|t|^\alpha)$. To avoid using facts that we have not proved, we will obtain our conclusions from Theorem 5.4.11. It is not hard to use that form of the criterion in this case, since

$$\begin{aligned} 1 - r\varphi(t) &\downarrow 1 - \exp(-|t|^\alpha) \text{ as } r \uparrow 1 \\ 1 - \exp(-|t|^\alpha) &\sim |t|^\alpha \text{ as } t \rightarrow 0 \end{aligned}$$

From this, it follows that the corresponding random walk is transient for $\alpha < 1$ and recurrent for $\alpha \geq 1$. The case $\alpha > 1$ is covered by Theorem 5.4.8, since these random walks have mean 0. The result for $\alpha = 1$ is new because the Cauchy distribution does not satisfy $S_n/n \rightarrow 0$ in probability. The random walks with $\alpha < 1$ are interesting because Exercise 5.4.1 implies

$$-\infty = \liminf S_n < \limsup S_n = \infty$$

but $P(|S_n| < M \text{ i.o.}) = 0$ for any $M < \infty$.

Remark The stable law examples are misleading in one respect. Shepp (1964) has proved that recurrent random walks may have arbitrarily large tails. To be precise, given a function $\epsilon(x) \downarrow 0$ as $x \uparrow \infty$, there is a recurrent random walk with $P(|X_1| \geq x) \geq \epsilon(x)$ for large x .

$d = 2$. Let $\alpha < 2$, and let $\varphi(t) = \exp(-|t|^\alpha)$, where $|t| = (t_1^2 + t_2^2)^{1/2}$. φ is the characteristic function of a random vector (X_1, X_2) that has two nice properties:

- (i) the distribution of (X_1, X_2) is invariant under rotations,
- (ii) X_1 and X_2 have symmetric stable laws with index α .

Again, $1 - r\varphi(t) \downarrow 1 - \exp(-|t|^\alpha)$ as $r \uparrow 1$ and $1 - \exp(-|t|^\alpha) \sim |t|^\alpha$ as $t \rightarrow 0$. Changing to polar coordinates and noticing

$$2\pi \int_0^\delta dx x x^{-\alpha} < \infty$$

when $1 - \alpha > -1$ shows the random walks with ch.f. $\exp(-|t|^\alpha)$, $\alpha < 2$ are transient. When $p < \alpha$, we have $E|X_1|^p < \infty$ by Exercise 3.8.4, so these examples show that Theorem 5.4.9 is reasonably sharp.

$d \geq 3$. The integral $\int_0^\delta dx x^{d-1} x^{-2} < \infty$, so if a random walk is recurrent in $d \geq 3$, its ch.f. must $\rightarrow 1$ faster than t^2 . In Exercise 3.3.16, we observed that (in one dimension) if $\varphi(r) = 1 + o(r^2)$, then $\varphi(r) \equiv 1$. By considering $\varphi(r\theta)$, where r is real and θ is a fixed vector, the last conclusion generalizes easily to $\mathbf{R}^d$, $d > 1$ and suggests that once we exclude walks that stay on a plane through 0, no three-dimensional random walks are recurrent.

A random walk in $\mathbf{R}^3$ is **truly three-dimensional** if the distribution of X_1 has $P(X_1 \cdot \theta \neq 0) > 0$ for all $\theta \neq 0$.

Theorem 5.4.14 *No truly three-dimensional random walk is recurrent.*

Proof We will deduce the result from Theorem 5.4.11. We begin with some arithmetic. If z is complex, the conjugate of $1 - z$ is $1 - \bar{z}$, so

$$\frac{1}{1 - z} = \frac{1 - \bar{z}}{|1 - z|^2} \quad \text{and} \quad \operatorname{Re} \frac{1}{1 - z} = \frac{\operatorname{Re}(1 - \bar{z})}{|1 - z|^2}$$

If $z = a + bi$ with $a \leq 1$, then using the previous formula and dropping the b^2 from the denominator

$$\operatorname{Re} \frac{1}{1-z} = \frac{1-a}{(1-a)^2 + b^2} \leq \frac{1}{1-a}$$

Taking $z = r\phi(t)$ and supposing for the second inequality that $0 \leq \operatorname{Re} \phi(t) \leq 1$, we have

$$(a) \quad \operatorname{Re} \frac{1}{1-r\phi(t)} \leq \frac{1}{\operatorname{Re}(1-r\phi(t))} \leq \frac{1}{\operatorname{Re}(1-\phi(t))}$$

The last calculation shows that it is enough to estimate

$$\operatorname{Re}(1-\phi(t)) = \int \{1 - \cos(x \cdot t)\} \mu(dx) \geq \int_{|x \cdot t| < \pi/3} \frac{|x \cdot t|^2}{4} \mu(dx)$$

by Lemma 5.4.13. Writing $t = \rho\theta$, where $\theta \in S = \{x : |x| = 1\}$ gives

$$(b) \quad \operatorname{Re}(1-\phi(\rho\theta)) \geq \frac{\rho^2}{4} \int_{|x \cdot \theta| < \pi/3\rho} |x \cdot \theta|^2 \mu(dx)$$

Fatou's lemma implies that if we let $\rho \rightarrow 0$ and $\theta(\rho) \rightarrow \theta$, then

$$(c) \quad \liminf_{\rho \rightarrow 0} \int_{|x \cdot \theta(\rho)| < \pi/3\rho} |x \cdot \theta(\rho)|^2 \mu(dx) \geq \int |x \cdot \theta|^2 \mu(dx) > 0$$

I claim this implies that for $\rho < \rho_0$

$$(d) \quad \inf_{\theta \in S} \int_{|x \cdot \theta| < \pi/3\rho} |x \cdot \theta|^2 \mu(dx) = C > 0$$

To get the last conclusion, observe that if it is false, then for $\rho = 1/n$ there is a θ_n so that

$$\int_{|x \cdot \theta_n| < n\pi/3} |x \cdot \theta_n|^2 \mu(dx) \leq 1/n$$

All the θ_n lie in S , a compact set, so if we pick a convergent subsequence we contradict (c). Combining (b) and (d) gives

$$\operatorname{Re}(1-\phi(\rho\theta)) \geq C\rho^2/4$$

Using the last result and (a) then changing to polar coordinates, we see that if δ is small (so $\operatorname{Re} \phi(y) \geq 0$ on $(-\delta, \delta)^d$)

$$\begin{aligned} \int_{(-\delta, \delta)^d} \operatorname{Re} \frac{1}{1-r\phi(y)} dy &\leq \int_0^{\delta\sqrt{d}} d\rho \rho^{d-1} \int d\theta \frac{1}{\operatorname{Re}(1-\phi(\rho\theta))} \\ &\leq C' \int_0^1 d\rho \rho^{d-3} < \infty \end{aligned}$$

when $d > 2$, so the desired result follows from Theorem 5.4.11. $\square$

Remark The analysis becomes much simpler when we consider random walks on $\mathbf{Z}^d$. The inversion formula given in Exercise 3.3.2 implies

$$P(S_n = 0) = (2\pi)^{-d} \int_{(-\pi, \pi)^d} \varphi^n(t) dt$$

Multiplying by r^n and summing gives

$$\sum_{n=0}^{\infty} r^n P(S_n = 0) = (2\pi)^{-d} \int_{(-\pi, \pi)^d} \frac{1}{1 - r\phi(t)} dt$$

In the case of simple random walk in $d = 3$, $\phi(t) = \frac{1}{3} \sum_{j=1}^3 \cos t_j$ is real.

$$\begin{aligned} \frac{1}{1 - r\phi(t)} &\uparrow \frac{1}{1 - \phi(t)} \quad \text{when } \phi(t) > 0 \\ 0 &\leq \frac{1}{1 - r\phi(t)} \leq 1 \quad \text{when } \phi(t) \leq 0 \end{aligned}$$

So, using the monotone and bounded convergence theorems

$$\sum_{n=0}^{\infty} P(S_n = 0) = (2\pi)^{-3} \int_{(-\pi, \pi)^3} \left(1 - \frac{1}{3} \sum_{i=1}^3 \cos x_i\right)^{-1} dx$$

This integral was first evaluated by Watson in 1939 in terms of elliptic integrals, which could be found in tables. Glasser and Zucker (1977) showed that it was

$$(\sqrt{6}/32\pi^3) \Gamma(1/24) \Gamma(5/24) \Gamma(7/24) \Gamma(11/24) = 1.51638606 \dots$$

so it follows from (5.3.1) that if $\beta_3 = P_0(T_0 < \infty)$, then

$$\sum_{n=0}^{\infty} P_0(S_n = 0) = \frac{1}{1 - \beta_3}$$

and hence $\beta_3 = 0.34053733 \dots$ For numerical results in $4 \leq d \leq 9$, see Kondo and Hara (1987).

Exercises

5.4.1 For a random walk on $\mathbf{R}$, there are only four possibilities, one of which has probability one.

- (i) $S_n = 0$ for all n .
- (ii) $S_n \rightarrow \infty$.
- (iii) $S_n \rightarrow -\infty$.
- (iv) $-\infty = \liminf S_n < \limsup S_n = \infty$.

5.4.2 **Ladder variables.** Let $\alpha(\omega) = \inf\{n : \omega_1 + \dots + \omega_n > 0\}$, where $\inf \emptyset = \infty$, and set $\alpha(\Delta) = \infty$. Let $\alpha_0 = 0$ and let

$$\alpha_k(\omega) = \alpha_{k-1}(\omega) + \alpha(\theta^{\alpha_{k-1}} \omega)$$

for $k \geq 1$. At time α_k , the random walk is at a record high value.

- (i) If $P(\alpha < \infty) < 1$, then $P(\sup S_n < \infty) = 1$.
- (ii) If $P(\alpha < \infty) = 1$, then $P(\sup S_n = \infty) = 1$.

5.4.3 Suppose f is superharmonic on $\mathbf{R}^d$, i.e.,

$$f(x) \geq \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) dy$$

Let $\xi_1, \xi_2, \dots$ be i.i.d. uniform on $B(0, 1)$, and define S_n by $S_n = S_{n-1} + \xi_n$ for $n \geq 1$ and $S_0 = x$. (i) Show that $X_n = f(S_n)$ is a supermartingale. (ii) Use this to conclude that in $d \leq 2$, nonnegative superharmonic functions must be constant. The example $f(x) = |x|^{2-d}$ shows this is false in $d > 2$.

5.4.4 Suppose h is harmonic on $\mathbf{R}^d$, i.e.,

$$h(x) = \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) dy$$

Let $\xi_1, \xi_2, \dots$ be i.i.d. uniform on $B(0, 1)$, and define S_n by $S_n = S_{n-1} + \xi_n$ for $n \geq 1$ and $S_0 = x$. (i) Show that $X_n = h(S_n)$ is a martingale. (ii) Use this to conclude that in any dimension, bounded harmonic functions must be constant. The Hewitt-Savage 0-1 law will be useful here.

5.5 Stationary Measures

A measure μ is said to be a **stationary measure** if

$$\sum_x \mu(x) p(x, y) = \mu(y)$$

The last equation says $P_\mu(X_1 = y) = \mu(y)$. Using the Markov property and induction, it follows that $P_\mu(X_n = y) = \mu(y)$ for all $n \geq 1$. If μ is a probability measure, we call μ a **stationary distribution**, and it represents a possible equilibrium for the chain. That is, if X_0 has distribution μ , then so does X_n for all $n \geq 1$. If we stretch our imagination a little, we can also apply this interpretation when μ is an infinite measure. (When the total mass is finite, we can divide by $\mu(S)$ to get a stationary distribution.) Before getting into the theory, we consider some examples.

Example 5.5.1 (Random walk) $S = \mathbf{Z}^d$. $p(x, y) = f(y - x)$, where $f(z) \geq 0$ and $\sum f(z) = 1$. In this case, $\mu(x) \equiv 1$ is a stationary measure, since

$$\sum_x p(x, y) = \sum_x f(y - x) = 1$$

A transition probability that has $\sum_x p(x, y) = 1$ is called **doubly stochastic**. This is obviously a necessary and sufficient condition for $\mu(x) \equiv 1$ to be a stationary measure.

Example 5.5.2 (Asymmetric simple random walk) $S = \mathbf{Z}$.

$$p(x, x+1) = p \quad p(x, x-1) = q = 1 - p$$

By the last example, $\mu(x) \equiv 1$ is a stationary measure. When $p \neq q$, $\mu(x) = (p/q)^x$ is a second one. To check this, we observe that

$$\begin{aligned} \sum_x \mu(x) p(x, y) &= \mu(y+1) p(y+1, y) + \mu(y-1) p(y-1, y) \\ &= (p/q)^{y+1} q + (p/q)^{y-1} p = (p/q)^y [p + q] = (p/q)^y \end{aligned}$$

To simplify the computations in the next two examples, we introduce a concept that is stronger than being a stationary measure. μ satisfies the **detailed balance condition** if

$$\mu(x)p(x, y) = \mu(y)p(y, x) \quad (5.5.1)$$

Summing over x gives

$$\sum_x \mu(x)p(x, y) = \mu(y)$$

(5.5.1) asserts that the amount of mass that moves from x to y in one jump is exactly the same as the amount that moves from y to x . A measure μ that satisfies (5.5.1) is said to be a **reversible measure**. Theorem 5.5.5 will explain the term reversible.

Example 5.5.3 (The Ehrenfest chain) $S = \{0, 1, \dots, r\}$

$$p(k, k+1) = (r-k)/r \quad p(k, k-1) = k/r$$

In this case, $\mu(x) = 2^{-r} \binom{r}{x}$ is a stationary distribution. One can check this without pencil and paper by observing that μ corresponds to flipping r coins to determine which urn each ball is to be placed in, and the transitions of the chain correspond to picking a coin at random and turning it over. Alternatively, you can check that

$$\begin{aligned} \mu(k+1)p(k+1, k) &= 2^{-r} \frac{r!}{(k+1)!(r-k-1)!} \cdot \frac{k+1}{r} = \frac{(r-1)!}{k!(r-k-1)!} \\ &= 2^{-r} \frac{r!}{k!(r-k)!} \cdot \frac{r-k}{r} = \mu(k-1)p(k-1, k) = \mu(k). \end{aligned}$$

Example 5.5.4 (Birth and death chains) $S = \{0, 1, 2, \dots\}$

$$p(x, x+1) = p_x \quad p(x, x) = r_x \quad p(x, x-1) = q_x$$

with $q_0 = 0$ and $p(i, j) = 0$ otherwise. In this case, there is the measure

$$\mu(x) = \prod_{k=1}^x \frac{p_{k-1}}{q_k}$$

which satisfies detailed balance:

$$\mu(x)p(x, x+1) = p_x \prod_{k=1}^x \frac{p_{k-1}}{q_k} = \mu(x+1)p(x+1, x)$$

The next result explains the name “reversible.”

Theorem 5.5.5 *Let μ be a stationary measure and suppose X_0 has “distribution” μ . Then $Y_m = X_{n-m}$, $0 \leq m \leq n$ is a Markov chain with initial measure μ and transition probability*

$$q(x, y) = \mu(y)p(y, x)/\mu(x)$$

q is called the **dual transition probability**. If μ is a reversible measure, then $q = p$.

Proof

$$\begin{aligned}
 P(Y_{m+1} = y | Y_m = x) &= P(X_{n-m-1} = y | X_{n-m} = x) \\
 &= \frac{P(X_{n-m-1} = y, X_{n-m} = x)}{P(X_{n-m} = x)} \\
 &= \frac{P(X_{n-m} = x | X_{n-m-1} = y) P(X_{n-m-1} = y)}{P(X_{n-m} = x)} \\
 &= \frac{\mu(y) p(y, x)}{\mu(x)}
 \end{aligned}$$

which proves the result. $\square$

While many chains have reversible measure, most do not. If there is a pair of states with $p(x, y) > 0$ and $p(y, x) = 0$, then it is impossible to have $\mu(p(x, y) = \pi(y) p(y, x))$. The $M/G/1$ queue also has this problem. The next result gives a necessary and sufficient condition for a chain to have a reversible measure.

Theorem 5.5.6 (Kolmogorov's cycle condition) *Suppose p is irreducible. A necessary and sufficient condition for the existence of a reversible measure is that (i) $p(x, y) > 0$ implies $p(y, x) > 0$, and (ii) for any loop $x_0, x_1, \dots, x_n = x_0$ with $\prod_{1 \leq i \leq n} p(x_i, x_{i-1}) > 0$,*

$$\prod_{i=1}^n \frac{p(x_{i-1}, x_i)}{p(x_i, x_{i-1})} = 1$$

Proof To prove the necessity of this condition, we note that irreducibility implies that any stationary measure has $\mu(x) > 0$ for all x , so (5.5.1) implies (i) holds. To check (ii), note that (5.5.1) implies that for the sequences considered here

$$\prod_{i=1}^n \frac{p(x_{i-1}, x_i)}{p(x_i, x_{i-1})} = \prod_{i=1}^n \frac{\mu(x_i)}{\mu(x_{i-1})} = 1$$

To prove sufficiency, fix $a \in S$, set $\mu(a) = 1$, and if $x_0 = a, x_1, \dots, x_n = x$ is a sequence with $\prod_{1 \leq i \leq n} p(x_i, x_{i-1}) > 0$ (irreducibility implies such a sequence will exist), we let

$$\mu(x) = \prod_{i=1}^n \frac{p(x_{i-1}, x_i)}{p(x_i, x_{i-1})}$$

The cycle condition guarantees that the last definition is independent of the path. To check (5.5.1) now, observe that if $p(y, x) > 0$, then adding $x_{n+1} = y$ to the end of a path to x we have

$$\mu(x) \frac{p(x, y)}{p(y, x)} = \mu(y) \quad \square$$

Only special chains have reversible measures, but as the next result shows, many Markov chains have stationary measures.

Example 5.5.8 (Renewal chain) $S = \{0, 1, 2, \dots\}$, $f_k \geq 0$, and $\sum_{k=1}^{\infty} f_k = 1$.

$$\begin{aligned} p(0, j) &= f_{j+1} && \text{for } j \geq 0 \\ p(i, i-1) &= 1 && \text{for } i \geq 1 \\ p(i, j) &= 0 && \text{otherwise} \end{aligned}$$

To explain the definition, let $\xi_1, \xi_2, \dots$ be i.i.d. with $P(\xi_m = j) = f_j$, let $T_0 = i_0$, and for $k \geq 1$ let $T_k = T_{k-1} + \xi_k$. T_k is the time of the k th arrival in a renewal process that has its first arrival at time i_0 . Let

$$Y_m = \begin{cases} 1 & \text{if } m \in \{T_0, T_1, T_2, \dots\} \\ 0 & \text{otherwise} \end{cases}$$

and let $X_n = \inf\{m - n : m \geq n, Y_m = 1\}$. $Y_m = 1$ if a renewal occurs at time m , and X_n is the amount of time until the first renewal $\geq n$.

An example should help clarify the definition:

$$\begin{array}{cccccccccccccccc} Y_n & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \\ X_n & 3 & 2 & 1 & 0 & 2 & 1 & 0 & 0 & 4 & 3 & 2 & 1 & 0 \end{array}$$

It is clear that if $X_n = i > 0$, then $X_{n+1} = i - 1$. When $X_n = 0$, the next ξ will be equal to k with probability f_k . In this case, the next positive value of Y is at time $n + k$, so $X_{n+1} = k - 1$.

$P_0(T_0 < \infty) = 1$, so 0 is always recurrent. If $f_K > 0$ but $f_k = 0$ for $k > K$, the chain is not irreducible on $\{0, 1, 2, \dots\}$ but it is on $\{0, 1, \dots, K-1\}$. To compute the stationary distribution, we use the cycle trick starting at 0. The chain will hit 0 if and only if the first jump is to a point $k \geq j$, so $\mu_0(j) = P(\xi > j)$. $E_0 T_0 = \sum_{j=0}^{\infty} P(\xi > j) = E\xi$. If $E\xi < \infty$, then $\pi(j) = P(\xi > j)/E\xi$ is a stationary distribution.

Theorem 5.5.7 allows us to construct a stationary measure for each closed set of recurrent states. Conversely, we have:

Theorem 5.5.9 *If p is irreducible and recurrent (i.e., all states are) then the stationary measure is unique up to constant multiples.*

Proof Let ν be a stationary measure and let $a \in S$.

$$\nu(z) = \sum_y \nu(y)p(y, z) = \nu(a)p(a, z) + \sum_{y \neq a} \nu(y)p(y, z)$$

Using the last identity to replace $\nu(y)$ on the right-hand side,

$$\begin{aligned} \nu(z) &= \nu(a)p(a, z) + \sum_{y \neq a} \nu(a)p(a, y)p(y, z) \\ &\quad + \sum_{x \neq a} \sum_{y \neq a} \nu(x)p(x, y)p(y, z) \\ &= \nu(a)P_a(X_1 = z) + \nu(a)P_a(X_1 \neq a, X_2 = z) \\ &\quad + P_\nu(X_0 \neq a, X_1 \neq a, X_2 = z) \end{aligned}$$

Continuing in the obvious way, we get

$$\begin{aligned} v(z) &= v(a) \sum_{m=1}^n P_a(X_k \neq a, 1 \leq k < m, X_m = z) \\ &\quad + P_v(X_j \neq a, 0 \leq j < n, X_n = z) \end{aligned}$$

The last term is ≥ 0 . Letting $n \rightarrow \infty$ gives $v(z) \geq v(a)\mu_a(z)$, where μ_a is the measure defined in Theorem 5.5.7 for $x = a$.

It is tempting to claim that recurrence implies

$$P_v(X_j \neq a, 0 \leq j < n) \rightarrow 0$$

but v may be an infinite measure, so we need another approach. It follows from Theorem 5.5.7 that μ_a is a stationary measure with $\mu_a(a) = 1$. (Here we are summing from 1 to T rather than from 0 to $T - 1$.) To turn the $\geq$ in the last equation into $=$, we observe

$$v(a) = \sum_x v(x) p^n(x, a) \geq v(a) \sum_x \mu_a(x) p^n(x, a) = v(a) \mu_a(a) = v(a)$$

Since $v(x) \geq v(a)\mu_a(x)$ and the left- and right-hand sides are equal, we must have $v(x) = v(a)\mu_a(x)$ whenever $p^n(x, a) > 0$. Since p is irreducible, it follows that $v(x) = v(a)\mu_a(x)$ for all $x \in S$, and the proof is complete. $\square$

Theorems 5.5.7 and 5.5.9 make a good team. The first result gives us a formula for a stationary distribution we call μ_x , and the second shows it is unique up to constant multiples. Together they allow us to derive a lot of formulas.

Having examined the existence and uniqueness of stationary measures, we turn our attention now to **stationary distributions**, i.e., probability measures π with $\pi p = \pi$. Stationary measures may exist for transient chains, e.g., random walks in $d \geq 3$, but:

Theorem 5.5.10 *If there is a stationary distribution, then all states y that have $\pi(y) > 0$ are recurrent.*

Proof Since $\pi p^n = \pi$, Fubini's theorem implies

$$\sum_x \pi(x) \sum_{n=1}^{\infty} p^n(x, y) = \sum_{n=1}^{\infty} \pi(y) = \infty$$

when $\pi(y) > 0$. Using Theorem 5.3.1 now gives

$$\infty = \sum_x \pi(x) \frac{\rho_{xy}}{1 - \rho_{yy}} \leq \frac{1}{1 - \rho_{yy}}$$

since $\rho_{xy} \leq 1$ and π is a probability measure. So $\rho_{yy} = 1$. $\square$

Theorem 5.5.11 *If p is irreducible and has stationary distribution π , then*

$$\pi(x) = 1/E_x T_x$$

Remark Recycling Chung's quote regarding Theorem 4.6.9, we note that the proof will make $\pi(x) = 1/E_x T_x$ obvious, but it seems incredible that

$$\sum_x \frac{1}{E_x T_x} p(x, y) = \frac{1}{E_y T_y}$$

Proof Irreducibility implies $\pi(x) > 0$, so all states are recurrent by Theorem 5.5.10. From Theorem 5.5.7,

$$\mu_x(y) = \sum_{n=0}^{\infty} P_x(X_n = y, T_x > n)$$

defines a stationary measure with $\mu_x(x) = 1$, and Fubini's theorem implies

$$\sum_y \mu_x(y) = \sum_{n=0}^{\infty} P_x(T_x > n) = E_x T_x$$

By Theorem 5.5.9, the stationary measure is unique up to constant multiples, so $\pi(x) = \mu_x(x)/E_x T_x$. Since $\mu_x(x) = 1$ by definition, the desired result follows. $\square$

If a state x has $E_x T_x < \infty$, it is said to be **positive recurrent**. A recurrent state with $E_x T_x = \infty$ is said to be **null recurrent**. Theorem 5.6.1 will explain these names. The next result helps us identify positive recurrent states.

Theorem 5.5.12 *If p is irreducible, then the following are equivalent:*

- (i) *Some x is positive recurrent.*
- (ii) *There is a stationary distribution.*
- (iii) *All states are positive recurrent.*

This result shows that being positive recurrent is a **class property**. If it holds for one state in an irreducible set, then it is true for all.

Proof (i) implies (ii). If x is positive recurrent, then

$$\pi(y) = \sum_{n=0}^{\infty} P_x(X_n = y, T_x > n) / E_x T_x$$

defines a stationary distribution.

(ii) implies (iii). Theorem 5.5.11 implies $\pi(y) = 1/E_y T_y$, and irreducibility tells us $\pi(y) > 0$ for all y , so $E_y T_y < \infty$.

(iii) implies (i). Trivial. $\square$

Example 5.5.13 **Birth and death chains** have a stationary distribution if and only if

$$\sum_x \prod_{k=1}^x \frac{p_{k-1}}{q_k} < \infty$$

By Theorem 5.3.11, the chain is recurrent if and only if

$$\sum_{m=0}^{\infty} \prod_{j=1}^m \frac{q_j}{p_j} = \infty$$

Example 5.5.14 ($M/G/1$ queue) Let $\mu = \sum ka_k$ be the mean number of customers that arrive during one service time. In Example 5.3.7, we showed that the chain is recurrent if and only if $\mu \leq 1$. We will now show that the chain is positive recurrent if and only if $\mu < 1$. First, suppose that $\mu < 1$. When $X_n > 0$, the chain behaves like a random walk that has jumps with mean $\mu - 1$, so if $N = \inf\{n \geq 0 : X_n = 0\}$, then $X_{N \wedge n} - (\mu - 1)(N \wedge n)$ is a martingale. If $X_0 = x > 0$, then the martingale property implies

$$x = E_x X_{N \wedge n} + (1 - \mu)E_x(N \wedge n) \geq (1 - \mu)E_x(N \wedge n)$$

since $X_{N \wedge n} \geq 0$, and it follows that $E_x N \leq x/(1 - \mu)$.

To prove that there is equality, observe that X_n decreases by at most one each time and for $x \geq 1$, $E_x T_{x-1} = E_1 T_0$, so $E_x N = cx$. To identify the constant, observe that

$$E_1 N = 1 + \sum_{k=0}^{\infty} a_k E_k N$$

so $c = 1 + \mu c$ and $c = 1/(1 - \mu)$. If $X_0 = 0$ then $p(0, 0) = a_0 + a_1$ and $p(0, k - 1) = a_k$ for $k \geq 2$. By considering what happens on the first jump, we see that (the first term may look wrong, but recall $k - 1 = 0$ when $k = 1$)

$$E_0 T_0 = 1 + \sum_{k=1}^{\infty} a_k \frac{k-1}{1-\mu} = 1 + \frac{\mu - (1 - a_0)}{1 - \mu} = \frac{a_0}{1 - \mu} < \infty$$

This shows that the chain is positive recurrent if $\mu < 1$. To prove the converse, observe that the arguments above show that if $E_0 T_0 < \infty$, then $E_k N < \infty$ for all k , $E_k N = ck$, and $c = 1/(1 - \mu)$, which is impossible if $\mu \geq 1$.

The last result when combined with Theorem 5.5.7 and 5.5.11 allows us to conclude that the stationary distribution has $\pi(0) = (1 - \mu)/a_0$. This may not seem like much, but the equations in $\pi p = \pi$ are:

$$\begin{aligned}\pi(0) &= \pi(0)(a_0 + a_1) + \pi(1)a_0 \\ \pi(1) &= \pi(0)a_2 + \pi(1)a_1 + \pi(2)a_0 \\ \pi(2) &= \pi(0)a_3 + \pi(1)a_2 + \pi(2)a_1 + \pi(3)a_0\end{aligned}$$

or, in general, for $j \geq 1$

$$\pi(j) = \sum_{i=0}^{j+1} \pi(i)a_{j+1-i}$$

The equations have a “triangular” form, so knowing $\pi(0)$, we can solve for $\pi(1), \pi(2), \dots$. The first expression,

$$\pi(1) = \pi(0)(1 - (a_0 + a_1))/a_0$$

is simple, but the formulas get progressively messier, and there is no nice closed form solution.

To close the section, we will give a self-contained proof of:

Theorem 5.5.15 *If p is irreducible and has a stationary distribution π , then any other stationary measure is a multiple of π .*

Remark This result is a consequence of Theorems 5.5.10 and 5.5.9, but we find the method of proof amusing.

Proof Since p is irreducible, $\pi(x) > 0$ for all x . Let φ be a concave function that is bounded on $(0, \infty)$, e.g., $\varphi(x) = x/(x+1)$. Define the **entropy** of μ by

$$\mathcal{E}(\mu) = \sum_y \varphi\left(\frac{\mu(y)}{\pi(y)}\right) \pi(y)$$

The reason for the name will become clear during the proof.

$$\begin{aligned} \mathcal{E}(\mu p) &= \sum_y \varphi\left(\sum_x \frac{\mu(x)p(x,y)}{\pi(y)}\right) \pi(y) = \sum_y \varphi\left(\sum_x \frac{\mu(x)}{\pi(x)} \cdot \frac{\pi(x)p(x,y)}{\pi(y)}\right) \pi(y) \\ &\geq \sum_y \sum_x \varphi\left(\frac{\mu(x)}{\pi(x)}\right) \frac{\pi(x)p(x,y)}{\pi(y)} \pi(y) \end{aligned}$$

since φ is concave, and $v(x) = \pi(x)p(x,y)/\pi(y)$ is a probability distribution. Since the $\pi(y)$'s cancel and $\sum_y p(x,y) = 1$, the last expression $= \mathcal{E}(\mu)$, and we have shown $\mathcal{E}(\mu p) \geq \mathcal{E}(\mu)$, i.e., the entropy of an arbitrary initial measure μ is increased by an application of p .

If $p(x,y) > 0$ for all x and y , and $\mu p = \mu$, it follows that $\mu(x)/\pi(x)$ must be constant, for otherwise there would be strict inequality in the application of Jensen's inequality. To get from the last special case to the general result, observe that if p is irreducible

$$\bar{p}(x,y) = \sum_{n=1}^{\infty} 2^{-n} p^n(x,y) > 0 \quad \text{for all } x, y$$

and $\mu p = \mu$ implies $\mu \bar{p} = \mu$. □

Exercises

5.5.1 Find the stationary distribution for the Bernoulli-Laplace model of diffusion from Exercise 5.1.5.

5.5.2 Let $w_{xy} = P_x(T_y < T_x)$. Show that $\mu_x(y) = w_{xy}/w_{yx}$.

5.5.3 Show that if p is irreducible and recurrent, then

$$\mu_x(y)\mu_y(z) = \mu_x(z)$$

5.5.4 Suppose p is irreducible and positive recurrent. Then $E_x T_y < \infty$ for all x, y .

5.5.5 Suppose p is irreducible and has a stationary measure μ with $\sum_x \mu(x) = \infty$. Then p is not positive recurrent.

5.5.6 Use Theorems 5.5.7 and 5.5.9 to show that for simple random walk, (i) the expected number of visits to k between successive visits to 0 is 1 for all k , and (ii) if we start from k , the expected number of visits to k before hitting 0 is $2k$.

5.5.7 Let $\xi_1, \xi_2, \dots$ be i.i.d. with $P(\xi_m = k) = a_{k+1}$ for $k \geq -1$, let $S_n = x + \xi_1 + \dots + \xi_n$, where $x \geq 0$, and let

$$X_n = S_n + \left(\min_{m \leq n} S_m \right)^-$$

(5.1.1) shows that X_n has the same distribution as the $M/G/1$ queue starting from $X_0 = x$. Use this representation to conclude that if $\mu = \sum k a_k < 1$, then as $n \rightarrow \infty$

$$\frac{1}{n} |\{m \leq n : X_{m-1} = 0, \xi_m = -1\}| \rightarrow (1 - \mu) \quad \text{a.s.}$$

and hence $\pi(0) = (1 - \mu)/a_0$ as proved earlier.

5.5.8 **$M/M/\infty$ queue.** In this chain, introduced in Exercise 5.3.8,

$$X_{n+1} = \sum_{m=1}^{X_n} \xi_{n,m} + Y_{n+1}$$

where $\xi_{n,m}$ are i.i.d. Bernoulli with mean p and Y_{n+1} is an independent Poisson mean λ . Compute the distribution at time 1 when $X_0 = \text{Poisson}(\mu)$ and use the result to find the stationary distribution.

5.5.9 Let $X_n \geq 0$ be a Markov chain and suppose $E_x X_1 \leq x - \epsilon$ for $x > K$, where $\epsilon > 0$. Let $Y_n = X_n + n\epsilon$ and $\tau = \inf\{n : X_n \leq K\}$. $Y_{n \wedge \tau}$ is a positive supermartingale and the optional stopping theorem implies $E_x \tau \leq x/\epsilon$. With assumptions about the behavior of the chain starting from points $x \leq K$, this leads to conditions for positive recurrence

5.6 Asymptotic Behavior

The first topic in this section is to investigate the asymptotic behavior of $p^n(x, y)$. If y is transient, $\sum_n p^n(x, y) < \infty$, so $p^n(x, y) \rightarrow 0$ as $n \rightarrow \infty$. To deal with the recurrent states, we let

$$N_n(y) = \sum_{m=1}^n 1_{\{X_m=y\}}$$

be the number of visits to y by time n .

Theorem 5.6.1 Suppose y is recurrent. For any $x \in S$, as $n \rightarrow \infty$

$$\frac{N_n(y)}{n} \rightarrow \frac{1}{E_y T_y} 1_{\{T_y < \infty\}} \quad P_x\text{-a.s.}$$

Here $1/\infty = 0$.

Proof Suppose first that we start at y . Let $R(k) = \min\{n \geq 1 : N_n(y) = k\}$ = the time of the k th return to y . Let $t_k = R(k) - R(k-1)$, where $R(0) = 0$. Since we have assumed $X_0 = y$, $t_1, t_2, \dots$ are i.i.d. and the strong law of large numbers implies

$$R(k)/k \rightarrow E_y T_y \quad P_y\text{-a.s.}$$

Since $R(N_n(y)) \leq n < R(N_n(y) + 1)$,

$$\frac{R(N_n(y))}{N_n(y)} \leq \frac{n}{N_n(y)} < \frac{R(N_n(y) + 1)}{N_n(y) + 1} \cdot \frac{N_n(y) + 1}{N_n(y)}$$

Letting $n \rightarrow \infty$, and recalling $N_n(y) \rightarrow \infty$ a.s. since y is recurrent, we have

$$\frac{n}{N_n(y)} \rightarrow E_y T_y \quad P_y\text{-a.s.}$$

To generalize now to $x \neq y$, observe that if $T_y = \infty$, then $N_n(y) = 0$ for all n and hence

$$N_n(y)/n \rightarrow 0 \quad \text{on } \{T_y = \infty\}$$

The strong Markov property implies that conditional on $\{T_y < \infty\}$, $t_2, t_3, \dots$ are i.i.d. and have $P_x(t_k = n) = P_y(T_y = n)$, so

$$R(k)/k = t_1/k + (t_2 + \dots + t_k)/k \rightarrow 0 + E_y T_y \quad P_x\text{-a.s.}$$

Repeating the proof for the case $x = y$ shows

$$N_n(y)/n \rightarrow 1/E_y T_y \quad P_x\text{-a.s. on } \{T_y < \infty\}$$

and combining this with the result for $\{T_y = \infty\}$ completes the proof. $\square$

Remark Theorem 5.6.1 should help explain the terms positive and null recurrent. If we start from x , then in the first case the asymptotic fraction of time spent at x is positive and in the second case it is 0.

Since $0 \leq N_n(y)/n \leq 1$, it follows from the bounded convergence theorem that $E_x N_n(y)/n \rightarrow E_x(1_{\{T_y < \infty\}}/E_y T_y)$, so

$$\frac{1}{n} \sum_{m=1}^n p^m(x, y) \rightarrow \rho_{xy}/E_y T_y \quad (5.6.1)$$

The last result was proved for recurrent y but also holds for transient y , since in that case, $E_y T_y = \infty$, and the limit is 0, since $\sum_m p^m(x, y) < \infty$.

(5.6.1) shows that the sequence $p^n(x, y)$ always converges in the Cesaro sense. The next example shows that $p^n(x, y)$ need not converge.

Example 5.6.2

$$p = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad p^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad p^3 = p, \quad p^4 = p^2, \dots$$

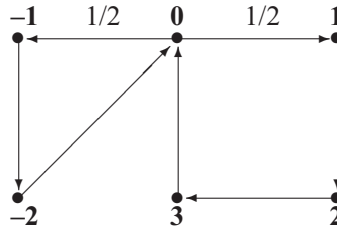
A similar problem also occurs in the Ehrenfest chain. In that case, if X_0 is even, then X_1 is odd, X_2 is even, $\dots$ so $p^n(x, x) = 0$ unless n is even. It is easy to construct examples with $p^n(x, x) = 0$ unless n is a multiple of 3 or 17 or $\dots$

Theorem 5.6.6 will show that this “periodicity” is the only thing that can prevent the convergence of the $p^n(x, y)$. First, we need a definition and two preliminary results. Let x be a recurrent state, let $I_x = \{n \geq 1 : p^n(x, x) > 0\}$, and let d_x be the greatest common divisor of I_x . d_x is called the **period** of x .

Example 5.6.3 (Triangle and square) Consider the transition matrix:

	-2	-1	0	1	2	3
-2	0	0	1	0	0	0
-1	1	0	0	0	0	0
0	0	0.5	0	0.5	0	0
1	0	0	0	0	1	0
2	0	0	0	0	0	1
3	0	0	1	0	0	0

In words, from 0 we are equally likely to go to 1 or -1. From -1 we go with probability one to -2 and then back to 0, from 1 we go to 2 then to 3 and back to 0. The name refers to the fact that $0 \rightarrow -1 \rightarrow -2 \rightarrow 0$ is a triangle and $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$ is a square.



Clearly, $p^3(0, 0) > 0$ and $p^4(0, 0) > 0$, so $3, 4 \in I_0$ and hence $d_0 = 1$.

The next result says that the period is a class property. In particular, we can use it to conclude that in the last example all states have period 1.

Lemma 5.6.4 *If $\rho_{xy} > 0$, then $d_y = d_x$.*

Proof Let K and L be such that $p^K(x, y) > 0$ and $p^L(y, x) > 0$. (x is recurrent, so $\rho_{yx} > 0$.)

$$p^{K+L}(y, y) \geq p^L(y, x)p^K(x, y) > 0$$

so d_y divides $K + L$, abbreviated $d_y | (K + L)$. Let n be such that $p^n(x, x) > 0$.

$$p^{K+n+L}(y, y) \geq p^L(y, x)p^n(x, x)p^K(x, y) > 0$$

so $d_y | (K + n + L)$, and hence $d_y | n$. Since $n \in I_x$ is arbitrary, $d_y | d_x$. Interchanging the roles of y and x gives $d_x | d_y$, and hence $d_x = d_y$. $\square$

If a chain is irreducible and $d_x = 1$, it is said to be aperiodic. The easiest way to check this is to find a state with $p(x, x) > 0$. The $M/G/1$ queue has $a_k > 0$ for all $k \geq 0$, so it has this property. The renewal chain is aperiodic if $\text{g.c.d.}\{k : f_k > 0\} = 1$.

Lemma 5.6.5 *If $d_x = 1$, then $p^m(x, x) > 0$ for $m \geq m_0$.*

Proof by example Consider the renewal chain (Example 5.5.8) with $f_5 = f_{12} = 1/2$. $5, 12 \in I_0$. Since $m, n \in I_0$ implies $m + n \in I_0$

$$I_0 = \{5, 10, 12, 15, 17, 20, 22, 24, 25, 27, 29, 30, 32, \\ 34, 35, 36, 37, 39, 40, 41, 42, 43, \dots\}$$

To check this, note that 5 gives rise to $10=5+5$ and $17=5+12$, 10 to 15 and 22, 12 to 17 and 24, etc. Once we have five consecutive numbers in I_0 , here 39–43, we have all the rest.

Proof We begin by observing that it enough to show that I_x will contain two consecutive integers: k and $k+1$. For then it will contain $2k, 2k+1, 2k+2, 3k, 3k+1, 3k+2, 3k+3$, and so on until we have k consecutive integers and then we $(k-1)k, (k-1)k+1, \dots (k-1)k+k-1$ and we have the rest.

To show that there are two consecutive integers, we cheat and use a fact from number theory: if the greatest common divisor of a set I_x is 1, then there are integers $i_1, \dots, i_m \in I_x$ and (positive or negative) integer coefficients c_i so that $c_1 i_1 + \dots + c_m i_m = 1$. Let $a_i = c_i^+$ and $b_i = (c_i)^-$. In words, the a_i are the positive coefficients and the b_i are -1 times the negative coefficients. Rearranging the last equation gives

$$a_1 i_1 + \dots + a_m i_m = (b_1 i_1 + \dots + b_m i_m) + 1$$

and we have found our two consecutive integers in I_x . In the numerical example $5 \cdot 5 - 2 \cdot 12 = 1$, which gives us consecutive integers 24 and 25. $\square$

Theorem 5.6.6 (Convergence theorem) Suppose p is irreducible, aperiodic (i.e., all states have $d_x = 1$), and has stationary distribution π . Then, as $n \rightarrow \infty$, $p^n(x, y) \rightarrow \pi(y)$.

Proof Let $S^2 = S \times S$. Define a transition probability $\bar{p}$ on $S \times S$ by

$$\bar{p}((x_1, y_1), (x_2, y_2)) = p(x_1, x_2)p(y_1, y_2)$$

i.e., each coordinate moves independently. Our first step is to check that $\bar{p}$ is irreducible. This may seem like a silly thing to do first, but this is the only step that requires aperiodicity. Since p is irreducible, there are K, L , so that $p^K(x_1, x_2) > 0$ and $p^L(y_1, y_2) > 0$. From Lemma 5.6.5 it follows that if M is large, $p^{L+M}(x_2, x_2) > 0$ and $p^{K+M}(y_2, y_2) > 0$, so

$$\bar{p}^{K+L+M}((x_1, y_1), (x_2, y_2)) > 0$$

Our second step is to observe that since the two coordinates are independent, $\bar{\pi}(a, b) = \pi(a)\pi(b)$ defines a stationary distribution for $\bar{p}$, and Theorem 5.5.10 implies that for $\bar{p}$ all states are recurrent. Let (X_n, Y_n) denote the chain on $S \times S$, and let T be the first time that this chain hits the diagonal $\{(y, y) : y \in S\}$. Let $T_{(x, x)}$ be the hitting time of (x, x) . Since $\bar{p}$ is irreducible and recurrent, $T_{(x, x)} < \infty$ a.s. and hence $T < \infty$ a.s. The final step is to observe that on $\{T \leq n\}$, the two coordinates X_n and Y_n have the same distribution. By considering the time and place of the first intersection and then using the Markov property,

$$\begin{aligned} P(X_n = y, T \leq n) &= \sum_{m=1}^n \sum_x P(T = m, X_m = x, X_n = y) \\ &= \sum_{m=1}^n \sum_x P(T = m, X_m = x) P(X_n = y | X_m = x) \\ &= \sum_{m=1}^n \sum_x P(T = m, Y_m = x) P(Y_n = y | Y_m = x) \\ &= P(Y_n = y, T \leq n) \end{aligned}$$

To finish up, we observe that

$$\begin{aligned} P(X_n = y) &= P(Y_n = y, T \leq n) + P(X_n = y, T > n) \\ &\leq P(Y_n = y) + P(X_n = y, T > n) \end{aligned}$$

and similarly, $P(Y_n = y) \leq P(X_n = y) + P(Y_n = y, T > n)$. So

$$|P(X_n = y) - P(Y_n = y)| \leq P(X_n = y, T > n) + P(Y_n = y, T > n)$$

and summing over y gives

$$\sum_y |P(X_n = y) - P(Y_n = y)| \leq 2P(T > n)$$

If we let $X_0 = x$ and let Y_0 have the stationary distribution π , then Y_n has distribution π , and it follows that

$$\sum_y |p^n(x, y) - \pi(y)| \leq 2P(T > n) \rightarrow 0$$

proving the desired result. If we recall the definition of the total variation distance given in Section 3.6, the last conclusion can be written as

$$\|p^n(x, \cdot) - \pi(\cdot)\| \leq P(T > n) \rightarrow 0 \quad \square$$

At first glance, it may seem strange to prove the convergence theorem by running independent copies of the chain. An approach that is slightly more complicated but explains better what is happening is to define

$$q((x_1, y_1), (x_2, y_2)) = \begin{cases} p(x_1, x_2)p(y_1, y_2) & \text{if } x_1 \neq y_1 \\ p(x_1, x_2) & \text{if } x_1 = y_1, x_2 = y_2 \\ 0 & \text{otherwise} \end{cases}$$

In words, the two coordinates move independently until they hit and then move together. It is easy to see from the definition that each coordinate is a copy of the original process. If T' is the hitting time of the diagonal for the new chain (X'_n, Y'_n) , then $X'_n = Y'_n$ on $T' \leq n$, so it is clear that

$$\sum_y |P(X'_n = y) - P(Y'_n = y)| \leq 2 P(X'_n \neq Y'_n) = 2P(T' > n)$$

On the other hand, T and T' have the same distribution, so $P(T' > n) \rightarrow 0$, and the conclusion follows as before. The technique used in the last proof is called **coupling**. Generally, this term refers to building two sequences X_n and Y_n on the same space to conclude that X_n converges in distribution by showing $P(X_n \neq Y_n) \rightarrow 0$, or more generally, that for some metric ρ , $\rho(X_n, Y_n) \rightarrow 0$ in probability.

Exercises

5.6.1 Suppose $S = \{0, 1\}$ and

$$p = \begin{pmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{pmatrix}$$

Use induction to show that

$$P_\mu(X_n = 0) = \frac{\beta}{\alpha + \beta} + (1 - \alpha - \beta)^n \left\{ \mu(0) - \frac{\beta}{\alpha + \beta} \right\}$$

5.6.2 Show that if S is finite and p is irreducible and aperiodic, then there is an m so that $p^m(x, y) > 0$ for all x, y .

5.6.3 Show that if S is finite, p is irreducible and aperiodic, and T is the coupling time defined in the proof of Theorem 5.6.6, then $P(T > n) \leq Cr^n$ for some $r < 1$ and $C < \infty$. So the convergence to equilibrium occurs exponentially rapidly in this case. Hint: First, consider the case in which $p(x, y) > 0$ for all x and y and reduce the general case to this one by looking at a power of p .

5.6.4 For any transition matrix p , define

$$\alpha_n = \sup_{i, j} \frac{1}{2} \sum_k |p^n(i, k) - p^n(j, k)|$$

The $1/2$ is there because for any i and j we can define r.v.'s X and Y so that $P(X = k) = p^n(i, k)$, $P(Y = k) = p^n(j, k)$, and

$$P(X \neq Y) = (1/2) \sum_k |p^n(i, k) - p^n(j, k)|$$

Show that $\alpha_{m+n} \leq \alpha_n \alpha_m$. Here you may find the coupling interpretation may help you from getting lost in the algebra. Using Lemma 2.7.1, we can conclude that

$$\frac{1}{n} \log \alpha_n \rightarrow \inf_{m \geq 1} \frac{1}{m} \log \alpha_m$$

so if $\alpha_m < 1$ for some m , it approaches 0 exponentially fast.

5.6.5 **Strong law for additive functionals.** Suppose p is irreducible and has stationary distribution π . Let f be a function that has $\sum |f(y)|\pi(y) < \infty$. Let T_x^k be the time of the k th return to x . (i) Show that

$$V_k^f = f(X(T_x^k)) + \cdots + f(X(T_x^{k+1} - 1)), \quad k \geq 1 \text{ are i.i.d.}$$

with $E|V_k^f| < \infty$. (ii) Let $K_n = \inf\{k : T_x^k \geq n\}$ and show that

$$\frac{1}{n} \sum_{m=1}^{K_n} V_m^f \rightarrow \frac{EV_1^f}{E_x T_x^1} = \sum_y f(y)\pi(y) \quad P_\mu - \text{a.s.}$$

(iii) Show that $\max_{1 \leq m \leq n} V_m^f / n \rightarrow 0$ and conclude

$$\frac{1}{n} \sum_{m=1}^n f(X_m) \rightarrow \sum_y f(y)\pi(y) \quad P_\mu - \text{a.s.}$$

for any initial distribution μ .

- 5.6.6 Central limit theorem for additive functionals.** Suppose in addition to the conditions in the Exercise 5.6.5 that $\sum f(y)\pi(y) = 0$, and $E_x(V_k^{f|})^2 < \infty$. (i) Use the random index central limit theorem (Exercise 3.4.6) to conclude that for any initial distribution μ

$$\frac{1}{\sqrt{n}} \sum_{m=1}^{K_n} V_m^f \Rightarrow c\chi \quad \text{under } P_\mu$$

- (ii) Show that $\max_{1 \leq m \leq n} V_m^{f|}/\sqrt{n} \rightarrow 0$ in probability and conclude

$$\frac{1}{\sqrt{n}} \sum_{m=1}^n f(X_m) \Rightarrow c\chi \quad \text{under } P_\mu$$

- 5.6.7 Ratio Limit Theorems.** Theorem 5.6.1 does not say much in the null recurrent case. To get a more informative limit theorem, suppose that y is recurrent and m is the (unique up to constant multiples) stationary measure on $C_y = \{z : \rho_{yz} > 0\}$. Let $N_n(z) = |\{m \leq n : X_m = z\}|$. Break up the path at successive returns to y and show that $N_n(z)/N_n(y) \rightarrow m(z)/m(y)$ P_x -a.s. for all $x, z \in C_y$. Note that $n \rightarrow N_n(z)$ is increasing, so this is much easier than the previous problem.

- 5.6.8** We got (5.6.1) from Theorem 5.6.1 by taking expected value. This does not work for the ratio in the previous exercise, so we need another approach. Suppose $z \neq y$. (i) Let $\bar{p}_n(x, z) = P_x(X_n = z, T_y > n)$ and decompose $p^m(x, z)$ according to the value of $J = \sup\{j \in [1, m] : X_j = y\}$ to get

$$\sum_{m=1}^n p^m(x, z) = \sum_{m=1}^n \bar{p}_m(x, z) + \sum_{j=1}^{n-1} p^j(x, y) \sum_{k=1}^{n-j} \bar{p}_k(y, z)$$

- (ii) Show that

$$\sum_{m=1}^n p^m(x, z) \bigg/ \sum_{m=1}^n p^m(x, y) \rightarrow \frac{m(z)}{m(y)}$$

5.7 Periodicity, Tail σ -Field*

Lemma 5.7.1 Suppose p is irreducible, recurrent, and all states have period d . Fix $x \in S$, and for each $y \in S$, let $K_y = \{n \geq 1 : p^n(x, y) > 0\}$. (i) There is an $r_y \in \{0, 1, \dots, d-1\}$ so that if $n \in K_y$, then $n = r_y \bmod d$, i.e., the difference $n - r_y$ is a multiple of d . (ii) Let $S_r = \{y : r_y = r\}$ for $0 \leq r < d$. If $y \in S_i$, $z \in S_j$, and $p^n(y, z) > 0$, then $n = (j - i) \bmod d$. (iii) $S_0, S_1, \dots, S_{d-1}$ are irreducible classes for p^d , and all states have period 1.

Proof (i) Let $m(y)$ be such that $p^{m(y)}(y, x) > 0$. If $n \in K_y$, then $p^{n+m(y)}(x, x)$ is positive, so $d|(n+m)$. Let $r_y = (d - m(y)) \bmod d$. (ii) Let m, n be such that $p^n(y, z), p^m(x, y) > 0$. Since $p^{n+m}(x, z) > 0$, it follows from (i) that $n+m = j \bmod d$. Since $m = i \bmod d$, the result follows. The irreducibility in (iii) follows immediately from (ii). The aperiodicity follows from the definition of the period as the g.c.d. $\{x : p^n(x, x) > 0\}$. $\square$

A partition of the state space $S_0, S_1, \dots, S_{d-1}$ satisfying (ii) in Lemma 5.7.1 is called a **cyclic decomposition** of the state space. Except for the choice of the set to put first, it is unique. (Pick an $x \in S$. It lies in some S_j , but once the value of j is known, irreducibility and (ii) allow us to calculate all the sets.)

Exercise 5.7.1 Find the decomposition for the Markov chain with transition probability

	1	2	3	4	5	6	7
1	0	0	0	.5	.5	0	0
2	.3	0	0	0	0	0	.7
3	0	0	0	0	0	0	1
4	0	0	1	0	0	0	0
5	0	0	1	0	0	0	0
6	0	1	0	0	0	0	0
7	0	0	0	.4	0	.6	0

Theorem 5.7.2 (Convergence theorem, periodic case) Suppose p is irreducible, has a stationary distribution π , and all states have period d . Let $x \in S$, and let $S_0, S_1, \dots, S_{d-1}$ be the cyclic decomposition of the state space with $x \in S_0$. If $y \in S_r$, then

$$\lim_{m \rightarrow \infty} p^{md+r}(x, y) = \pi(y)d$$

Proof If $y \in S_0$, then using (iii) in Lemma 5.7.1 and applying Theorem 5.6.6 to p^d shows

$$\lim_{m \rightarrow \infty} p^{md}(x, y) \text{ exists}$$

To identify the limit, we note that (5.6.1) implies

$$\frac{1}{n} \sum_{m=1}^n p^m(x, y) \rightarrow \pi(y)$$

and (ii) of Lemma 5.7.1 implies $p^m(x, y) = 0$ unless $d|m$, so the limit in the first display must be $\pi(y)d$. If $y \in S_r$ with $1 \leq r < d$, then

$$p^{md+r}(x, y) = \sum_{z \in S_r} p^r(x, z) p^{md}(z, y)$$

Since $y, z \in S_r$ it follows from the first case in the proof that $p^{md}(z, y) \rightarrow \pi(y)d$ as $m \rightarrow \infty$. $p^{md}(z, y) \leq 1$, and $\sum_z p^r(x, z) = 1$, so the result follows from the dominated convergence theorem. $\square$

Let $\mathcal{F}'_n = \sigma(X_{n+1}, X_{n+2}, \dots)$ and $\mathcal{T} = \cap_n \mathcal{F}'_n$ be the tail σ -field. The next result is due to Orey. The proof we give is from Blackwell and Freedman (1964).

Theorem 5.7.3 Suppose p is irreducible, recurrent, and all states have period d , $\mathcal{T} = \sigma(\{X_0 \in S_r\} : 0 \leq r < d)$.

Remark To be precise, if μ is any initial distribution and $A \in \mathcal{T}$, then there is an r so that $A = \{X_0 \in S_r\}$ P_μ -a.s.

Proof We build up to the general result in three steps.

Case 1. Suppose $P(X_0 = x) = 1$. Let $T_0 = 0$, and for $n \geq 1$, let $T_n = \inf\{m > T_{n-1} : X_m = x\}$ be the time of the n th return to x . Let

$$V_n = (X(T_{n-1}), \dots, X(T_n - 1))$$

The vectors V_n are i.i.d. by Exercise 5.3.1, and the tail σ -field is contained in the exchangeable field of the V_n , so the Hewitt-Savage 0-1 law (Theorem 2.5.4, proved there for r.v.'s taking values in a general measurable space) implies that $\mathcal{T}$ is trivial in this case.

Case 2. Suppose that the initial distribution is concentrated on one cyclic class, say S_0 . If $A \in \mathcal{T}$ then $P_x(A) \in \{0, 1\}$ for each x by case 1. If $P_x(A) = 0$ for all $x \in S_0$, then $P_\mu(A) = 0$. Suppose $P_y(A) > 0$, and hence $= 1$, for some $y \in S_0$. Let $z \in S_0$. Since p^d is irreducible and aperiodic on S_0 , there is an n so that $p^n(z, y) > 0$ and $p^n(y, y) > 0$. If we write $1_A = 1_B \circ \theta_n$ then the Markov property implies

$$1 = P_y(A) = E_y(E_y(1_B \circ \theta_n | \mathcal{F}_n)) = E_y(E_{X_n} 1_B)$$

so $P_y(B) = 1$. Another application of the Markov property gives

$$P_z(A) = E_z(E_{X_n} 1_B) \geq p^n(z, y) > 0$$

so $P_z(A) = 1$, and since $z \in S_0$ is arbitrary, $P_\mu(A) = 1$.

General Case. From case 2, we see that $P(A|X_0 = y) \equiv 1$ or $\equiv 0$ on each cyclic class. This implies that either $\{X_0 \in S_r\} \subset A$ or $\{X_0 \in S_r\} \cap A = \emptyset$ P_μ a.s. Conversely, it is clear that $\{X_0 \in S_r\} = \{X_{nd} \in S_r \text{ i.o.}\} \in \mathcal{T}$, and the proof is complete. $\square$

The next result will help us identify the tail σ -field in transient examples.

Theorem 5.7.4 Suppose X_0 has initial distribution μ . The equations

$$h(X_n, n) = E_\mu(Z | \mathcal{F}_n) \quad \text{and} \quad Z = \lim_{n \rightarrow \infty} h(X_n, n)$$

set up a 1-1 correspondence between bounded $Z \in \mathcal{T}$ and bounded **space-time harmonic functions**, i.e., bounded $h : S \times \{0, 1, \dots\} \rightarrow \mathbf{R}$, so that $h(X_n, n)$ is a martingale.

Proof Let $Z \in \mathcal{T}$, write $Z = Y_n \circ \theta_n$, and let $h(x, n) = E_x Y_n$.

$$E_\mu(Z | \mathcal{F}_n) = E_\mu(Y_n \circ \theta_n | \mathcal{F}_n) = h(X_n, n)$$

by the Markov property, so $h(X_n, n)$ is a martingale. Conversely, if $h(X_n, n)$ is a bounded martingale, using Theorems 4.2.11 and 4.6.7 shows $h(X_n, n) \rightarrow Z \in \mathcal{T}$ as $n \rightarrow \infty$, and $h(X_n, n) = E_\mu(Z | \mathcal{F}_n)$. $\square$

Exercise 5.7.2 A random variable Z with $Z = Z \circ \theta$, and hence $= Z \circ \theta_n$ for all n , is called **invariant**. Show there is a 1-1 correspondence between bounded invariant random variables and bounded harmonic functions. We will have more to say about invariant r.v.'s in Section 7.1.

Example 5.7.5 (Simple random walk in d dimensions) We begin by constructing a coupling for this process. Let $i_1, i_2, \dots$ be i.i.d. uniform on $\{1, \dots, d\}$. Let $\xi_1, \xi_2, \dots$ and

$\eta_1, \eta_2, \dots$ be i.i.d. uniform on $\{-1, 1\}$. Let e_j be the j th unit vector. Construct a coupled pair of d -dimensional simple random walks by

$$X_n = X_{n-1} + e(i_n)\xi_n$$

$$Y_n = \begin{cases} Y_{n-1} + e(i_n)\xi_n & \text{if } X_{n-1}^{i_n} = Y_{n-1}^{i_n} \\ Y_{n-1} + e(i_n)\eta_n & \text{if } X_{n-1}^{i_n} \neq Y_{n-1}^{i_n} \end{cases}$$

In words, the coordinate that changes is always the same in the two walks, and once they agree in one coordinate, future movements in that direction are the same. It is easy to see that if $X_0^i - Y_0^i$ is even for $1 \leq i \leq d$, then the two random walks will hit with probability one.

Let $L_0 = \{z \in \mathbf{Z}^d : z^1 + \dots + z^d \text{ is even}\}$ and $L_1 = \mathbf{Z}^d - L_0$. Although we have only defined the notion for the recurrent case, it should be clear that L_0, L_1 is the cyclic decomposition of the state space for simple random walk. If $S_n \in L_i$, then $S_{n+1} \in L_{1-i}$ and p^2 is irreducible on each L_i . To couple two random walks starting from $x, y \in L_i$, let them run independently until the first time all the coordinate differences are even, and then use the last coupling. In the remaining case, $x \in L_0, y \in L_1$ coupling is impossible.

The next result should explain our interest in coupling two d -dimensional simple random walks.

Theorem 5.7.6 *For d -dimensional simple random walk,*

$$\mathcal{T} = \sigma(\{X_0 \in L_i, i = 0, 1\})$$

Proof Let $x, y \in L_i$, and let X_n, Y_n be a realization of the coupling defined above for $X_0 = x$ and $Y_0 = y$. Let $h(x, n)$ be a bounded space-time harmonic function. The martingale property implies $h(x, 0) = E_x h(X_n, n)$. If $|h| \leq C$, it follows from the coupling that

$$|h(x, 0) - h(y, 0)| = |Eh(X_n, n) - Eh(Y_n, n)| \leq 2CP(X_n \neq Y_n) \rightarrow 0$$

so $h(x, 0)$ is constant on L_0 and L_1 . Applying the last result to $h'(x, m) = h(x, n + m)$, we see that $h(x, n) = a_n^i$ on L_i . The martingale property implies $a_n^i = a_{n+1}^{1-i}$, and the desired result follows from Theorem 5.7.4. $\square$

Example 5.7.7 (Ornstein's coupling) Let $p(x, y) = f(y - x)$ be the transition probability for an irreducible aperiodic random walk on $\mathbf{Z}$. To prove that the tail σ -field is trivial, pick M large enough so that the random walk generated by the probability distribution $f_M(x)$ with $f_M(x) = c_M f(x)$ for $|x| \leq M$ and $f_M(x) = 0$ for $|x| > M$ is irreducible and aperiodic. Let $Z_1, Z_2, \dots$ be i.i.d. with distribution f and let $W_1, W_2, \dots$ be i.i.d. with distribution f_M . Let $X_n = X_{n-1} + Z_n$ for $n \geq 1$. If $X_{n-1} = Y_{n-1}$, we set $X_n = Y_n$. Otherwise, we let

$$Y_n = \begin{cases} Y_{n-1} + Z_n & \text{if } |Z_n| > m \\ Y_{n-1} + W_n & \text{if } |Z_n| \leq m \end{cases}$$

In words, the big jumps are taken in parallel and the small jumps are independent. The recurrence of one-dimensional random walks with mean 0 implies $P(X_n \neq Y_n) \rightarrow 0$. Repeating the proof of Theorem 5.7.6, we see that $\mathcal{T}$ is trivial.

The tail σ -field in Theorem 5.7.6 is essentially the same as in Theorem 5.7.3. To get a more interesting $\mathcal{T}$, we look at:

Example 5.7.8 (Random walk on a tree) To facilitate definitions, we will consider the system as a random walk on a group with three generators a, b, c that have $a^2 = b^2 = c^2 = e$, the identity element. To form the random walk, let $\xi_1, \xi_2, \dots$ be i.i.d. with $P(\xi_n = x) = 1/3$ for $x = a, b, c$, and let $X_n = X_{n-1}\xi_n$. (This is equivalent to a random walk on the tree in which each vertex has degree 3 but the algebraic formulation is convenient for computations.) Let L_n be the length of the word X_n when it has been reduced as much as possible, with $L_n = 0$ if $X_n = e$. The reduction can be done as we go along. If the last letter of X_{n-1} is the same as ξ_n , we erase it, otherwise we add the new letter. It is easy to see that L_n is a Markov chain with a transition probability that has $p(0, 1) = 1$ and

$$p(j, j-1) = 1/3 \quad p(j, j+1) = 2/3 \quad \text{for } j \geq 1$$

As $n \rightarrow \infty$, $L_n \rightarrow \infty$. From this, it follows easily that the word X_n has a limit in the sense that the i th letter X_n^i stays the same for large n . Let X_∞ be the limiting word, i.e., $X_\infty^i = \lim X_n^i$. $\mathcal{T} \supset \sigma(X_\infty^i, i \geq 1)$, but it is easy to see that this is not all. If $S_0 =$ the words of even length, and $S_1 = S_0^c$, then $X_n \in S_i$ implies $X_{n+1} \in S_{1-i}$, so $\{X_0 \in S_0\} \in \mathcal{T}$. Can the reader prove that we have now found all of $\mathcal{T}$? As Fermat once said, “I have a proof but it won’t fit in the margin.”

Remark This time the solution does not involve elliptic curves but uses “ h -paths.” See Furstenberg (1970) or decode the following: “Condition on the exit point (the infinite word). Then the resulting RW is an h -process, which moves closer to the boundary with probability $2/3$ and farther with probability $1/3$ ($1/6$ each to the two possibilities). Two such random walks couple, provided they have same parity.” The quote is from Robin Pemantle, who says he consulted Itai Benjamini and Yuval Peres.

5.8 General State Space*

In this section, we will generalize the results from countable state space to a collection of Markov chains with uncountable state space called Harris chains. The developments here are motivated by three ideas. First, the proofs for countable state space if there is one point in the state space that the chain hits with probability one. (Think, for example, about the construction of the stationary measure via the cycle trick.) Second, a recurrent Harris chain can be modified to contain such a point. Third, the collection of Harris chains is a comfortable level of generality; broad enough to contain a large number of interesting examples, yet restrictive enough to allow for a rich theory.

We say that a Markov chain X_n is a **Harris chain** if we can find sets $A, B \in \mathcal{S}$, a function q with $q(x, y) \geq \epsilon > 0$ for $x \in A, y \in B$, and a probability measure ρ concentrated on B so that:

- (i) If $\tau_A = \inf\{n \geq 0 : X_n \in A\}$, then $P_z(\tau_A < \infty) > 0$ for all $z \in S$.
- (ii) If $x \in A$ and $C \subset B$, then $p(x, C) \geq \int_C q(x, y) \rho(dy)$.

To explain the definition we turn to some examples:

Example 5.8.1 (Countable state space) If S is countable and there is a point a with $\rho_{xa} > 0$ for all x (a condition slightly weaker than irreducibility), then we can take $A = \{a\}$, $B = \{b\}$, where b is any state with $p(a, b) > 0$, $\mu = \delta_b$ the point mass at b , and $q(a, b) = p(a, b)$.

Example 5.8.2 (Chains with continuous densities) Suppose $X_n \in \mathbf{R}^d$ is a Markov chain with a transition probability that has $p(x, dy) = p(x, y) dy$, where $(x, y) \rightarrow p(x, y)$ is continuous. Pick (x_0, y_0) so that $p(x_0, y_0) > 0$. Let A and B be open sets around x_0 and y_0 that are small enough so that $p(x, y) \geq \epsilon > 0$ on $A \times B$. If we let $\rho(C) = |B \cap C|/|B|$, where $|B|$ is the Lebesgue measure of B , then (ii) holds. If (i) holds, then X_n is a Harris chain.

For concrete examples, consider:

(a) **Diffusion processes** are a large class of examples that lie outside the scope of this book, but are too important to ignore. When things are nice, specifically, if the generator of X has Hölder continuous coefficients satisfying suitable growth conditions, see the Appendix of Dynkin (1965), then $P_x(X_1 \in dy) = p(x, y) dy$, and p satisfies the previously noted conditions.

(b) The **discrete Ornstein-Uhlenbeck process**. Let $\xi_1, \xi_2, \dots$ be i.i.d. standard normals and let $V_n = \theta V_{n-1} + \xi_n$. The Ornstein-Uhlenbeck process is a diffusion process $\{V_t, t \in [0, \infty)\}$ that models the velocity of a particle suspended in a liquid. See, e.g., Breiman (1968) section 16.1. Looking at V_t at integer times (and dividing by a constant to make the variance 1) gives a Markov chain with the indicated distributions.

Example 5.8.3 (GI/G/1 queue, or storage model) Let $\xi_1, \xi_2, \dots$ be i.i.d. and define W_n inductively by $W_n = (W_{n-1} + \xi_n)^+$. If $P(\xi_n < 0) > 0$, then we can take $A = B = \{0\}$ and (i) and (ii) hold. To explain the first name in the title, consider a queueing system in which customers arrive at times of a renewal process, i.e., at times $0 = T_0 < T_1 < T_2 \dots$ with $\zeta_n = T_n - T_{n-1}$, $n \geq 1$ i.i.d. Let η_n , $n \geq 0$, be the amount of service time the n th customer requires and let $\xi_n = \eta_{n-1} - \zeta_n$. I claim that W_n is the amount of time the n th customer has to wait to enter service. To see this, notice that the $(n-1)$ th customer adds η_{n-1} to the server's workload, and if the server is busy at all times in $[T_{n-1}, T_n)$, he reduces his workload by ζ_n . If $W_{n-1} + \eta_{n-1} < \zeta_n$, then the server has enough time to finish his work and the next arriving customer will find an empty queue.

The second name in the title refers to the fact that W_n can be used to model the contents of a storage facility. For an intuitive description, consider water reservoirs. We assume that rain storms occur at times of a renewal process $\{T_n : n \geq 1\}$, that the n th rainstorm contributes an amount of water η_n , and that water is consumed at constant rate c . If we let $\zeta_n = T_n - T_{n-1}$ as before, and $\xi_n = \eta_{n-1} - c\zeta_n$, then W_n gives the amount of water in the reservoir just before the n th rainstorm.

History Lesson. Doeblin was the first to prove results for Markov chains on general state space. He supposed that there was an n so that $p^n(x, C) \geq \epsilon \rho(C)$ for all $x \in S$ and $C \subset S$. See Doob (1953), section V.5, for an account of his results. Harris (1956) generalized Doeblin's result by observing that it was enough to have a set A so that (i) holds and the chain viewed on A ($Y_k = X(T_A^k)$, where $T_A^k = \inf\{n > T_A^{k-1} : X_n \in A\}$ and $T_A^0 = 0$) satisfies Doeblin's condition. Our formulation, as well as most of the proofs in this section,

follows Athreya and Ney (1978). For a nice description of the “traditional approach,” see Revuz (1984).

Given a Harris chain on $(S, \mathcal{S})$, we will construct a Markov chain $\bar{X}_n$ with transition probability $\bar{p}$ on $(\bar{S}, \bar{\mathcal{S}})$, where $\bar{S} = S \cup \{\alpha\}$ and $\bar{\mathcal{S}} = \{B, B \cup \{\alpha\} : B \in \mathcal{S}\}$. The aim, as advertised earlier, is to manufacture a point α that the process hits with probability 1 in the recurrent case.

$$\begin{aligned} \text{If } x \in S - A \quad & \bar{p}(x, C) = p(x, C) \text{ for } C \in \mathcal{S} \\ \text{If } x \in A \quad & \bar{p}(x, \{\alpha\}) = \epsilon \\ & \bar{p}(x, C) = p(x, C) - \epsilon \rho(C) \text{ for } C \in \mathcal{S} \\ \text{If } x = \alpha \quad & \bar{p}(\alpha, D) = \int \rho(dx) \bar{p}(x, D) \text{ for } D \in \bar{\mathcal{S}} \end{aligned}$$

Intuitively, $\bar{X}_n = \alpha$ corresponds to X_n being distributed on B according to ρ . Here, and in what follows, we will reserve A and B for the special sets that occur in the definition and use C and D for generic elements of $\mathcal{S}$. We will often simplify notation by writing $\bar{p}(x, \alpha)$ instead of $\bar{p}(x, \{\alpha\})$, $\mu(\alpha)$ instead of $\mu(\{\alpha\})$, etc.

Our next step is to prove three technical lemmas that will help us develop the theory. Define a transition probability ν by

$$\nu(x, \{x\}) = 1 \quad \text{if } x \in S \quad \nu(\alpha, C) = \rho(C)$$

In words, V leaves mass in S alone but returns the mass at α to S and distributes it according to ρ .

Lemma 5.8.4 $\nu\bar{p} = \bar{p}$ and $\bar{p}\nu = p$.

Proof Before giving the proof, we would like to remind the reader that measures multiply the transition probability on the left, i.e., in the first case we want to show $\mu\nu\bar{p} = \mu\bar{p}$. If we first make a transition according to ν and then one according to $\bar{p}$, this amounts to one transition according to $\bar{p}$, since only mass at α is affected by ν and

$$\bar{p}(\alpha, D) = \int \rho(dx) \bar{p}(x, D)$$

The second equality also follows easily from the definition. In words, if $\bar{p}$ acts first and then ν , then ν returns the mass at α to where it came from. $\square$

From Lemma 5.8.4, it follows easily that we have:

Lemma 5.8.5 *Let Y_n be an inhomogeneous Markov chain with $p_{2k} = \nu$ and $p_{2k+1} = \bar{p}$. Then $\bar{X}_n = Y_{2n}$ is a Markov chain with transition probability $\bar{p}$ and $X_n = Y_{2n+1}$ is a Markov chain with transition probability p .*

Lemma 5.8.5 shows that there is an intimate relationship between the asymptotic behavior of X_n and of $\bar{X}_n$. To quantify this, we need a definition. If f is a bounded measurable function on S , let $\bar{f} = \nu f$, i.e., $\bar{f}(x) = f(x)$ for $x \in S$ and $\bar{f}(\alpha) = \int f d\rho$.

Lemma 5.8.6 *If μ is a probability measure on $(S, \mathcal{S})$, then*

$$E_\mu f(X_n) = E_\mu \bar{f}(\bar{X}_n)$$

Proof Observe that if X_n and $\bar{X}_n$ are constructed as in Lemma 5.8.5, and $P(\bar{X}_0 \in S) = 1$, then $X_0 = \bar{X}_0$ and X_n is obtained from $\bar{X}_n$ by making a transition according to ν . $\square$

The last three lemmas will allow us to obtain results for X_n from those for $\bar{X}_n$. We turn now to the task of generalizing the results from countable state space to $\bar{X}_n$. Before developing the theory, we will give one last example that explains why some of the statements are messy.

Example 5.8.7 (Perverted O.U. process) Take the discrete O.U. process from part (b) of Example 5.8.2 and modify the transition probability at the integers $x \geq 2$ so that

$$\begin{aligned} p(x, \{x+1\}) &= 1 - x^{-2} \\ p(x, A) &= x^{-2}|A| \quad \text{for } A \subset (0, 1) \end{aligned}$$

p is the transition probability of a Harris chain, but

$$P_2(X_n = n+2 \text{ for all } n) > 0$$

I can sympathize with the reader who thinks that such chains will not arise “in applications,” but it seems easier (and better) to adapt the theory to include them than to modify the assumptions to exclude them.

5.8.1 Recurrence and Transience

We begin with the dichotomy between recurrence and transience. Let $R = \inf\{n \geq 1 : \bar{X}_n = \alpha\}$. If $P_\alpha(R < \infty) = 1$, then we call the chain **recurrent**, otherwise we call it **transient**. Let $R_1 = R$ and for $k \geq 2$, let $R_k = \inf\{n > R_{k-1} : \bar{X}_n = \alpha\}$ be the time of the k th return to α . The strong Markov property implies $P_\alpha(R_k < \infty) = P_\alpha(R < \infty)^k$, so $P_\alpha(\bar{X}_n = \alpha \text{ i.o.}) = 1$ in the recurrent case and $= 0$ in the transient case.

The next result generalizes Lemma 5.3.2. (If x is recurrent and $\rho_{xy} > 0$, then y is recurrent and $\rho_{yx} = 1$.)

Theorem 5.8.8 Let $\lambda(C) = \sum_{n=1}^{\infty} 2^{-n} \bar{p}^n(\alpha, C)$. In the recurrent case, if $\lambda(C) > 0$, then $P_\alpha(\bar{X}_n \in C \text{ i.o.}) = 1$. For λ -a.e. x , $P_x(R < \infty) = 1$.

Proof The first conclusion follows from Exercise 4.6.4. For the second let $D = \{x : P_x(R < \infty) < 1\}$ and observe that if $p^n(\alpha, D) > 0$ for some n , then

$$P_\alpha(\bar{X}_m = \alpha \text{ i.o.}) \leq \int \bar{p}^n(\alpha, dx) P_x(R < \infty) < 1 \quad \square$$

Remark Example 5.8.7 shows that we cannot expect to have $P_x(R < \infty) = 1$ for all x . To see that even when the state space is countable, we need not hit every point starting from α (see Exercise 5.8.2).

5.8.2 Stationary Measures

Theorem 5.8.9 In the recurrent case, there is a σ -finite stationary measure $\bar{\mu} \ll \lambda$.

Proof Let $R = \inf\{n \geq 1 : \bar{X}_n = \alpha\}$, and let

$$\bar{\mu}(C) = E_\alpha \left(\sum_{n=0}^{R-1} 1_{\{\bar{X}_n \in C\}} \right) = \sum_{n=0}^{\infty} P_\alpha(\bar{X}_n \in C, R > n)$$

Repeating the proof of Theorem 5.5.7 shows that $\bar{\mu}\bar{p} = \bar{\mu}$. If we let $\mu = \bar{\mu}\nu$, then it follows from Lemma 5.8.4 that $\bar{\mu}\nu p = \bar{\mu}\bar{p}\nu = \bar{\mu}\nu$, so $\mu p = \mu$.

To prove that it is σ -finite, let $G_{k,\delta} = \{x : \bar{p}^k(x, \alpha) \geq \delta\}$. Let $T_1 = \inf\{n : X(T_1) \in G_k\}$ and for $m \geq 2$, let $X(T_m) = \inf\{n \geq T_{m-1}\}$, and let $M = \sup\{m : T_m < R\}$. By assumption, $EM < 1/\delta$. In between T_m and T_{m-1} there can be at most k visits to $G_{k,\delta}$. The fact that $\bar{\mu} \ll \lambda$ is clear by comparing the definitions. $\square$

To investigate uniqueness of the stationary measure, we begin with:

Lemma 5.8.10 *If ν is a σ -finite stationary measure for p , then $\nu(A) < \infty$ and $\bar{\nu} = \nu\bar{p}$ is a stationary measure for $\bar{p}$ with $\bar{\nu}(\alpha) < \infty$.*

Proof We will first show that $\nu(A) < \infty$. If $\nu(A) = \infty$, then part (ii) of the definition implies $\nu(C) = \infty$ for all sets C with $\rho(C) > 0$. If $B = \cup_i B_i$ with $\nu(B_i) < \infty$, then $\rho(B_i) = 0$ by the last observation and $\rho(B) = 0$ by countable subadditivity, a contradiction. So $\nu(A) < \infty$ and $\bar{\nu}(\alpha) = \nu\bar{p}(\alpha) = \epsilon\nu(A) < \infty$. Using the fact that $\nu p = \nu$, we find

$$\nu\bar{p}(C) = \nu(C) - \epsilon\nu(A)\rho(B \cap C)$$

the last subtraction being well-defined since $\nu(A) < \infty$, and it follows that $\bar{\nu}\nu = \nu$. To check $\bar{\nu}\bar{p} = \bar{\nu}$, we observe that Lemma 5.8.4 and the last result imply $\bar{\nu}\bar{p} = \bar{\nu}\nu\bar{p} = \nu\bar{p} = \bar{\nu}$. $\square$

Theorem 5.8.11 *Suppose p is recurrent. If ν is a σ -finite stationary measure, then $\nu = \bar{\nu}(\alpha)\mu$, where μ is the measure constructed in the proof of Theorem 5.8.9.*

Proof By Lemma 5.8.10, it suffices to prove that if $\bar{\nu}$ is a stationary measure for $\bar{p}$ with $\bar{\nu}(\alpha) < \infty$, then $\bar{\nu} = \bar{\nu}(\alpha)\bar{\mu}$. Repeating the proof of Theorem 5.5.9 with $a = \alpha$, it is easy to show that $\bar{\nu}(C) \geq \bar{\nu}(\alpha)\bar{\mu}(C)$. Continuing to compute as in that proof:

$$\bar{\nu}(\alpha) = \int \bar{\nu}(dx) \bar{p}^n(x, \alpha) \geq \bar{\nu}(\alpha) \int \bar{\mu}(dx) \bar{p}^n(x, \alpha) = \bar{\nu}(\alpha)\bar{\mu}(\alpha) = \bar{\nu}(\alpha)$$

Let $S_n = \{x : p^n(x, \alpha) > 0\}$. By assumption, $\cup_n S_n = S$. If $\bar{\nu}(D) > \bar{\nu}(\alpha)\bar{\mu}(D)$ for some D , then $\bar{\nu}(D \cap S_n) > \bar{\nu}(\alpha)\bar{\mu}(D \cap S_n)$, and it follows that $\bar{\nu}(\alpha) > \bar{\nu}(\alpha)$ a contradiction. $\square$

5.8.3 Convergence Theorem

We say that a recurrent Harris chain X_n is **aperiodic** if $\text{g.c.d. } \{n \geq 1 : p^n(\alpha, \alpha) > 0\} = 1$. This occurs, for example, if we can take $A = B$ in the definition for then $p(\alpha, \alpha) > 0$.

Theorem 5.8.12 *Let X_n be an aperiodic recurrent Harris chain with stationary distribution π . If $P_x(R < \infty) = 1$, then as $n \rightarrow \infty$,*

$$\|p^n(x, \cdot) - \pi(\cdot)\| \rightarrow 0$$

Note. Here $\|\cdot\|$ denotes the total variation distance between the measures. Lemma 5.8.8 guarantees that π a.e. x satisfies the hypothesis.

Proof In view of Lemma 5.8.6, it suffices to prove the result for $\bar{p}$. We begin by observing that the existence of a stationary probability measure and the uniqueness result in Theorem 5.8.11 imply that the measure constructed in Theorem 5.8.9 has $E_\alpha R = \bar{\mu}(S) < \infty$. As in the proof of Theorem 5.6.6, we let X_n and Y_n be independent copies of the chain with initial distributions δ_x and π , respectively, and let $\tau = \inf\{n \geq 0 : X_n = Y_n = \alpha\}$. For $m \geq 0$, let S_m (resp. T_m) be the times at which X_n (resp. Y_n) visit α for the $(m+1)$ th time. $S_m - T_m$ is a random walk with mean 0 steps, so $M = \inf\{m \geq 1 : S_m = T_m\} < \infty$ a.s., and it follows that this is true for τ as well. The computations in the proof of Theorem 5.6.6 show $|P(X_n \in C) - P(Y_n \in C)| \leq P(\tau > n)$. Since this is true for all C , $\|p^n(x, \cdot) - \pi(\cdot)\| \leq P(\tau > n)$, and the proof is complete. $\square$

5.8.4 GI/G/1 Queue

For the rest of the section, we will concentrate on the GI/G/1 queue. Let $\xi_1, \xi_2, \dots$ be i.i.d., let $W_n = (W_{n-1} + \xi_n)^+$, and let $S_n = \xi_1 + \dots + \xi_n$. Recall $\xi_n = \eta_{n-1} - \zeta_n$, where the η 's are service times, ζ 's are the interarrival times, and suppose $E\xi_n < 0$ so that Exercise 5.8.5 implies there is a stationary distribution.

Explicit formulas for the distribution of M are in general difficult to obtain. However, this can be done if either the arrival or service distribution is exponential. One reason for this is:

Lemma 5.8.13 *Suppose $X, Y \geq 0$ are independent and $P(X > x) = e^{-\lambda x}$. Show that $P(X - Y > x) = ae^{-\lambda x}$, where $a = P(X - Y > 0)$.*

Proof Let F be the distribution of Y

$$\begin{aligned} P(X - Y > x) &= \int_0^\infty dF(y) e^{-\lambda(x+y)} \\ &= e^{-\lambda x} \int_0^\infty dF(y) e^{-\lambda y} = e^{-\lambda x} P(X - Y > 0) \end{aligned}$$

where the last equality follows from setting $x = 0$ in the first one. $\square$

Example 5.8.14 (Exponential service time) Suppose $P(\eta_n > x) = e^{-\beta x}$ and $E\zeta_n > E\eta_n$. Let $T = \inf\{n : S_n > 0\}$ and $L = S_T$, setting $L = -\infty$ if $T = \infty$. The lack of memory property of the exponential distribution implies that $P(L > x) = re^{-\beta x}$, where $r = P(T < \infty)$. To compute the distribution of the maximum, M , let $T_1 = T$ and let $T_k = \inf\{n > T_{k-1} : S_n > S_{T_{k-1}}\}$ for $k \geq 2$. The strong Markov property implies that if $T_k < \infty$, then $S(T_{k+1}) - S(T_k) =_d L$ and is independent of $S(T_k)$. Using this and breaking things down according to the value of $K = \inf\{k : L_{k+1} = -\infty\}$, we see that for $x > 0$ the density function

$$P(M = x) = \sum_{k=1}^{\infty} r^k (1-r) e^{-\beta x} \beta^k x^{k-1} / (k-1)! = \beta r (1-r) e^{-\beta x (1-r)}$$

To complete the calculation, we need to calculate r . To do this, let

$$\phi(\theta) = E \exp(\theta \xi_n) = E \exp(\theta \eta_{n-1}) E \exp(-\theta \zeta_n)$$

which is finite for $0 < \theta < \beta$, since $\zeta_n \geq 0$ and η_{n-1} has an exponential distribution. It is easy to see that

$$\phi'(0) = E\xi_n < 0 \quad \lim_{\theta \uparrow \beta} \phi(\theta) = \infty$$

so there is a $\theta \in (0, \beta)$ with $\phi(\theta) = 1$. Example 4.2.3 implies $\exp(\theta S_n)$ is a martingale. Theorem 4.4.1 implies $1 = E \exp(\theta S_{T \wedge n})$. Letting $n \rightarrow \infty$ and noting that $(S_n | T = n)$ has an exponential distribution and $S_n \rightarrow -\infty$ on $\{T = \infty\}$, we have

$$1 = r \int_0^\infty e^{\theta x} \beta e^{-\beta x} dx = \frac{r\beta}{\beta - \theta}$$

Example 5.8.15 (Poisson arrivals) Suppose $P(\zeta_n > x) = e^{-\alpha x}$ and $E\zeta_n > E\eta_n$. Let $\bar{S}_n = -S_n$. Reversing time as in (ii) of Exercise 5.8.6, we see (for $n \geq 1$)

$$P\left(\max_{0 \leq k < n} \bar{S}_k < \bar{S}_n \in A\right) = P\left(\min_{1 \leq k \leq n} \bar{S}_k > 0, \bar{S}_n \in A\right)$$

Let $\psi_n(A)$ be the common value of the last two expressions and let $\psi(A) = \sum_{n \geq 0} \psi_n(A)$. $\psi_n(A)$ is the probability the random walk reaches a new maximum (or ladder height, see Exercise 5.4.2 in A at time n , so $\psi(A)$ is the number of ladder points in A with $\psi(\{0\}) = 1$. Letting the random walk take one more step

$$P\left(\min_{1 \leq k \leq n} \bar{S}_k > 0, \bar{S}_{n+1} \leq x\right) = \int F(x - z) d\psi_n(z)$$

The last identity is valid for $n = 0$ if we interpret the left-hand side as $F(x)$. Let $\tau = \inf\{n \geq 1 : \bar{S}_n \leq 0\}$ and $x \leq 0$. Integrating by parts on the right-hand side and then summing over $n \geq 0$ gives

$$\begin{aligned} P(\bar{S}_\tau \leq x) &= \sum_{n=0}^{\infty} P\left(\min_{1 \leq k \leq n} \bar{S}_k > 0, \bar{S}_{n+1} \leq x\right) \\ &= \int_{y \leq x} \psi[0, x - y] dF(y) \end{aligned} \quad (5.8.1)$$

The limit $y \leq x$ comes from the fact that $\psi((-\infty, 0)) = 0$.

Let $\bar{\xi}_n = \bar{S}_n - \bar{S}_{n-1} = -\xi_n$. Exercise 5.8.13 implies $P(\bar{\xi}_n > x) = ae^{-\alpha x}$. Let $\bar{T} = \inf\{n : \bar{S}_n > 0\}$. $E\bar{\xi}_n > 0$, so $P(\bar{T} < \infty) = 1$. Let $J = \bar{S}_T$. As in the previous example, $P(J > x) = e^{-\alpha x}$. Let $V_n = J_1 + \cdots + J_n$. V_n is a rate α Poisson process, so $\psi[0, x - y] = 1 + \alpha(x - y)$ for $x - y \geq 0$. Using (5.8.1) now and integrating by parts gives

$$\begin{aligned} P(\bar{S}_\tau \leq x) &= \int_{y \leq x} (1 + \alpha(x - y)) dF(y) \\ &= F(x) + \alpha \int_{-\infty}^x F(y) dy \quad \text{for } x \leq 0 \end{aligned} \quad (5.8.2)$$

Since $P(\bar{S}_n = 0) = 0$ for $n \geq 1$, $-\bar{S}_\tau$ has the same distribution as S_T , where $T = \inf\{n : S_n > 0\}$. Combining this with part (ii) of Exercise 5.8.6 gives a “formula” for $P(M > x)$. Straightforward but somewhat tedious calculations show that if $B(s) = E \exp(-s\eta_n)$, then

$$E \exp(-sM) = \frac{(1 - \alpha \cdot E\eta)s}{s - \alpha + \alpha B(s)}$$

a result known as the **Pollaczek-Khintchine formula**. The computations we omitted can be found in Billingsley (1979) on p. 277 or several times in Feller, Vol. II (1971).

Exercises

- 5.8.1 $\bar{X}_n$ is recurrent if and only if $\sum_{n=1}^{\infty} \bar{p}^n(\alpha, \alpha) = \infty$.
- 5.8.2 If X_n is a recurrent Harris chain on a countable state space, then S can only have one irreducible set of recurrent states but may have a nonempty set of transient states. For a concrete example, consider a branching process in which the probability of no children $p_0 > 0$ and set $A = B = \{0\}$.
- 5.8.3 Use Exercise 5.8.1 and imitate the proof of Theorem 5.5.10 to show that a Harris chain with a stationary distribution must be recurrent.
- 5.8.4 Suppose X_n is a recurrent Harris chain. Show that if (A', B') is another pair satisfying the conditions of the definition, then Theorem 5.8.8 implies $P_\alpha(\bar{X}_n \in A' \text{ i.o.}) = 1$, so the recurrence or transience does not depend on the choice of (A, B) .
- 5.8.5 In the $GI/G/1$ queue, the waiting time W_n and the random walk $S_n = X_0 + \xi_1 + \cdots + \xi_n$ agree until $N = \inf\{n : S_n < 0\}$, and at this time $W_N = 0$. Use this observation as we did in Example 5.3.7 to show that Example 5.8.3 is transient when $E\xi_n > 0$, recurrent when $E\xi_n \leq 0$, and there is a stationary distribution when $E\xi_n < 0$.
- 5.8.6 Let $m_n = \min(S_0, S_1, \dots, S_n)$, where S_n is the random walk defined previously.
 (i) Show that $S_n - m_n =_d W_n$. (ii) Let $\xi'_m = \xi_{n+1-m}$ for $1 \leq m \leq n$. Show that $S_n - m_n = \max(S'_0, S'_1, \dots, S'_n)$. (iii) Conclude that as $n \rightarrow \infty$ we have $W_n \Rightarrow M \equiv \max(S'_0, S'_1, S'_2, \dots)$.
- 5.8.7 In the discrete O.U. process, X_{n+1} is normal with mean θX_n and variance 1. What happens to the recurrence and transience if instead Y_{n+1} is normal with mean 0 and variance $\beta^2|Y_n|$?